

Industrial Attachment at



Akij Food & Beverage Limited

Krishnopura, Barobarra, Dhamrai

Date: 15th May, 2024 to 12th June 2024

IPE 4209: Industrial Attachment

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January 04, 2025

ACKNOWLEDGEMENT

Apart from our efforts, the success of any project depends largely on the encouragement and guidelines of many individuals. We take this opportunity to express our gratitude to the people who have been instrumental in completing this project. First of all, we are very thankful to our Honorable Director Professor Dr. Ashequl Alam Rana sir, and Assistant Professor Raihan Ahmed Joy Sir, Head of the Department of Industrial & Production Engineering (IPE) for granting us of this attachment.

We want to show the most significant appreciation to our respected supervisor Md. Kamrul Hasansir. We cannot thank him enough for all his tremendous support and help. He inspired us greatly to work on this project. His willingness to motivate us contributed tremendously to our project.

Grateful appreciations to all honorable teachers of **“National Institute of Textile Engineering & Research (NITER)”** for their kind help and assistance in various ways during the course of the study.

Another person who really helped us to get these big opportunities is Md. Toyeb Ali sir, Assistant General Manager (Head of production) of AFBL for guiding us to do our internship in Akij Food & Beverage Ltd. We acknowledge those honorable ‘Respondents’ who have provided the information for this study uncovering many personal matters before an outsider like us. We are greatly in debt with them.

We would also like to thank Sk. Shamim Uddin Sir, Chairman of AFBL. It allowed us to participate and learn about the many aspects of the Company mentioned above. Besides, we would like to thank Google for providing us with helpful information and facilities to complete this venture. The guidance and support received from the members who contributed to this project were vital for the project's success.

Special thanks also to Md Abdur Rahman Sir, Executive (HRM) who guided us to complete our internship.

Lastly, we would like to express our heartfelt gratitude to our parents and other members of the industry for their blessings, encouragement and constant inspiration throughout our academic career. Lastly, thanks to everyone who helped us in this whole Industrial Attachment.

EXECUTIVE SUMMARY

AFBL is one of the leading business institutions of Bangladesh. It's contributing directly to the economy of Bangladesh through its co-operation with customers and export earnings. One of the significant organizational goals is to reduce poverty and hunger through large-scale employment generation. The journey of AFBL started in 2006. Its evolution to one of the top food manufacturing companies has set the grounds for development for any potentialities.

The report focuses on the product portfolio where we found AFBL takes an extensive way to all forms of agro-processed food products, considering all methods of ensuring hygienic and quality food products and their comprehensive product range. We also tried to write about the CSR activities of AFBL as the time to think only about profit has gone. A new era has come to think about the people living around us AFBL group gives CSR effort on revolving around the four premises environment, energy preservation, community & people to improve the quality of livelihood. Moreover, AFBL gives strengths and competencies developed over the years; AFBL aims to provide the best working environment, good quality products for its customer. In addition, the significant strengths of AFBL in Bangladesh experience, large economy, more significant sources of finance, greater control over sources of raw materials, risk-pooling, colossal distribution network, conforming to standards, economies of scale, creating a potential market in the new world place. Finally, the food processing industry has contributed to economic progress in Bangladesh. In today's world, the export of food is ultimately an essential way of improving Bangladesh's economy.

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CHAPTER 1

DESCRIPTION OF ORGANIZATION

1.1 Akij Ventures Limited - A Bangladeshi Industrial Powerhouse:

One of the largest industrial conglomerates in Bangladesh is the Akij Group. Textile, tobacco, food & beverage, cement, ceramics, printing and packaging, pharmaceuticals, and consumer products are among the industries covered by this Cluster. Akij Group paid EUR 390 million in taxes, making it the largest local taxpayer and contributing two percent of this country's total budget for 2009. Akij also provides services in the field of healthcare, information, and communication technologies. In 2009, its turnover reached 89 billion Taka.

Founded in the 1940s by industrialist Sheikh Akij Uddin as a jute trading business, Akij Group later expanded into cigarettes and various other sectors. Despite its growth and contributions, the group has faced criticism for employing child labor.

The legacy of AKIJ GROUP is over half a century old and over the years Akij has established itself as the full of confidence and much revered industrial family of Bangladesh. In Bangladesh, the Akij Group is one of the biggest conglomerates. It's made up of 24 big issues, involving various activities and different products. It has been over 50 years since the Akij Group began its business as a small trader of jute. Since then, in the industrial area of Bangladesh, progress has been rapid. The group employs a large number of people and cares for them as members of the Akij family. Supporting the country's development and social welfare is also a matter of concern for NGOs.

- Akij Food & Beverage Ltd.
- Akij Ceramics Company Ltd
- Akij Match Factory Ltd.
- Akij Ceramics Ltd.
- Akij Flour Mill Industry
- Akij Cement Company Ltd.
- Akij Jute Mills Ltd.
- Akij Textile Mills Ltd.
- Akij Rice Mill Industry
- Akij Gas Company Ltd. etc.

1.2 About Akij Food and Beverage

Akij Food & Beverage Limited (AFBL) has started commercial operation in 2000. AFBL is now No.1 beverage company in Bangladesh with a strong brand presence. AFBL has 1300+

Distributors throughout Bangladesh with 7000+ direct employees and 25000+ indirect employees. AFBL has a factory on 65-acre land at Dhamrai, Dhaka, Bangladesh. AFBL delivers the best carbonated beverages, juice, fruit drinks, and snacks, etc.

Table 1.1: Description Table

Name	Akij Food & Beverage LTD. (AFBL)
Logo	
Type	Business Enterprise, Manufacturer
Year of Establishment	2006
Location & Address	Dhaka Office: Akij House,198, Bir Uttam Showkat Sarak, Dhaka-1208 Factory Address: Krisnopura, Dhamrai, Dhaka-1350
Number of Employees	5000+
Product Manufactured	Carbonated Soft Drinks, Mineral Water, Fruit Juice, Snacks & Dairy Products.
Export Inception	1981
Annual Production Capacity	Soft Drinks 3-7Million (approx.) Mineral Water 3-5Million (approx.) Fruit Juice & others 7-8Million (approx.)
Annual Sell Range (USD)	5 -10 million (approx.)
Number of Building	8 (approx.)
Certificate	ISO 9001:2015, BSTI, Halal Food, GMP, HACCP WCS,

1.3 Mission & Vision:

To be the market chief in quality nourishment and refreshment and fulfilling the longing and need of the network working in concordance with our clients, representatives and colleagues. Also, the most respected food & beverage company in Bangladesh by commitment to quality.

1.4 Physical Infrastructure:

AFBL is a constructed well-furnished business factory consisting of (8) different main buildings including different department.

1. CS Building
2. Production Building
3. Store & Distribution
4. Utility Building
5. PPS plant
6. Recycle & Bricks Plant

Besides these buildings as a physical structure, they have 2 canteens for general stuffs & administrative personals, dormitory building & check points.

Detailed structure of the main buildings is given below:

- CS (Common Service)/Admin Building:
 - Office Room
 - Conference Room
 - Quality Control Room
 - Medical
 - Security
- Production Building:
 - CSD (Soft drinks)
 - Aseptic production (fruit-juice)
 - Akij Dairy Plant
 - DPW-2 Plant
 - Snacks Plant
- Utility & Chemical Building
 - Maintenance Room
 - ETP, WTP
 - Chemical Storage
 - Power House

- Boiler (Fire & Water tube)

1.5 Additional facilities in AFBL:

- **Optimized Layout:** The physical layout of the factory is well-organized, promoting worker discipline and facilitating smooth production flow.
- **Advanced Technology Integration:** AFBL leverages cutting-edge machinery and technology to achieve optimal performance. This includes automation for material handling, which minimizes waste and maximizes productivity.
- **Real-Time Production Monitoring:** A sophisticated system provides instant data access for manufacturing, enabling rapid decision-making and a robust quality control process.
- **Skilled Workforce Utilization:** Manpower allocation is optimized for maximum efficiency. This ensures that the right skills are employed in the most impactful manner.
- **Performance-Based Incentive System:** AFBL motivates its employees through an incentive program that rewards exceptional performance, fostering a culture of continuous improvement and high-quality production.
- **Upskilling and Development Programs:** A dedicated training center offers ongoing education and skill development opportunities for the workforce, ensuring they remain current with industry best practices.

Dynamic Production Planning: Engineers continuously analyze production methods and implement the most efficient processes to maintain high production rates and meet evolving needs.

CHAPTER 2

ORGANIZATION OVERVIEW

2.1 Human Capital - The Cornerstone of AFBL's Success:

Effective manpower management, also known as Human Resource Management (HRM), involves strategically placing the right people with the right skills in the right roles at the right time to achieve organizational goals. The success of an organization depends a great deal on human resources. An unskilled workforce cannot be compensated for by investments in other assets. Therefore, organizations have to be even more careful about managing their human resources than they are with the selection of fixed assets.

Investing in Employee Potential

AFBL fosters a work environment that values creativity and innovation. This approach motivates employees and empowers them to contribute their best work. By investing in employee development, AFBL ensures its workforce possesses the skills and knowledge necessary to excel in their roles and contribute to the company's continued success.

2.2 Functions of Man Power Management (HRM)

- Recruitment policy and procedure¹¹
- Conformation policy and procedure
- Placement of employees
- Human Resource Development
- Organizational structure review and modification
- Job description preparation
- General services
- Safety-security
- Welfare activities for employees

2.3 HRM Department in AFBL

- Ten departments are working in AFBL to achieve their sustainable competitive advantage. Ten departments of AFBL are- The brand department, HR Department, Trade Department, Creative department, Digital Department, Media Department, Export-import Department, Research and Development Department, Finance Department, and Event Department.

- The Department of Brand has joined the work of the Media and Digital Office. All three divisions are in charge of building brand situating and mark value in the market.
- At least one brand executive shall be assigned to each brand. The center strategic marketing The Director of Marketing of AFBL and the Assistant General shall make decisions. Manager alongside a Brand supervisor. All the marketing is performed by senior executives Their advertising effort to their well-known brands. The vast majority of the marketing effort incorporates Sponsorship events, Sales advancement and promotion, Gift hamper dissemination, Branding for T-shirts, and mugs, Avertisements on TV RADIO, and NEWSPAPER, and more.
- The business and sales distribution drive of AFBL is isolated into 27 Areas across the nation Employee Engagement
- The Sales Department handles sales activities. It is directed by a few Senior Sales Admins.
- The Financial Year of this organization begins in July and finishes in one year from now June. The determined deals, target, yearly spending plan, business improvement arranging, and marking goals all begin toward the start of the Fiscal year.HR Analytics and Technology
- For each of the marketing promotional activities particular budget is requested and this budget needs to be approved by the managers and director.
- AFBL has an inside-structured and well-equipped Design Department. The Department ofCreative Design makes the structure for all sorts of advancements, similar to plans, and logos for any promotional activities. Additionally for any brand advancement for Printing, Billboard, Newspaper magazine ads, and so forth.
- The internal Research Department (R&D) conducts various questionnaire surveys, interviews, and qualitative and quantitative research for product development.

All the authority inside correspondence of AFBL and Akij Group is done through email and IP telephone. Akij Group itself has tremendous arrangement of system server associations throughout the workplace. An IT office run by IT specialists is in charge of all the upkeep of this system and securities. Every one of the workers has their Workstation to server and IP telephones. These advances work much more effectively and quicker

In order to make HRM more strategic in AFBL, management has reduced most of the administrative and paperwork from HRM that might hold HRM back. **AFBL** has organized HRM function as follows:

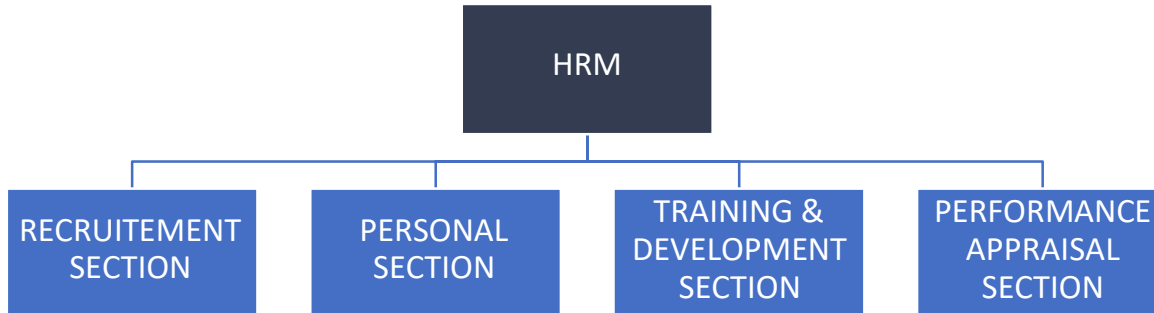


Figure 2.1: HRM Function

2.4 R&D Department in AFBL

The term research and development (R&D) is used to describe a series of activities that companies undertake to innovate and introduce new products and services. R&D is often the first stage in the development process.

Mainly R&D works with systematic activity with combining applied research. Also, implementing new idea with new scope and hope.

In AFBL, R&D department is most important things because of their responsibilities to;

- Unique product launch
- Cost reducing
- Taste
- Nutrition
- Ensure food safety.
- Product development

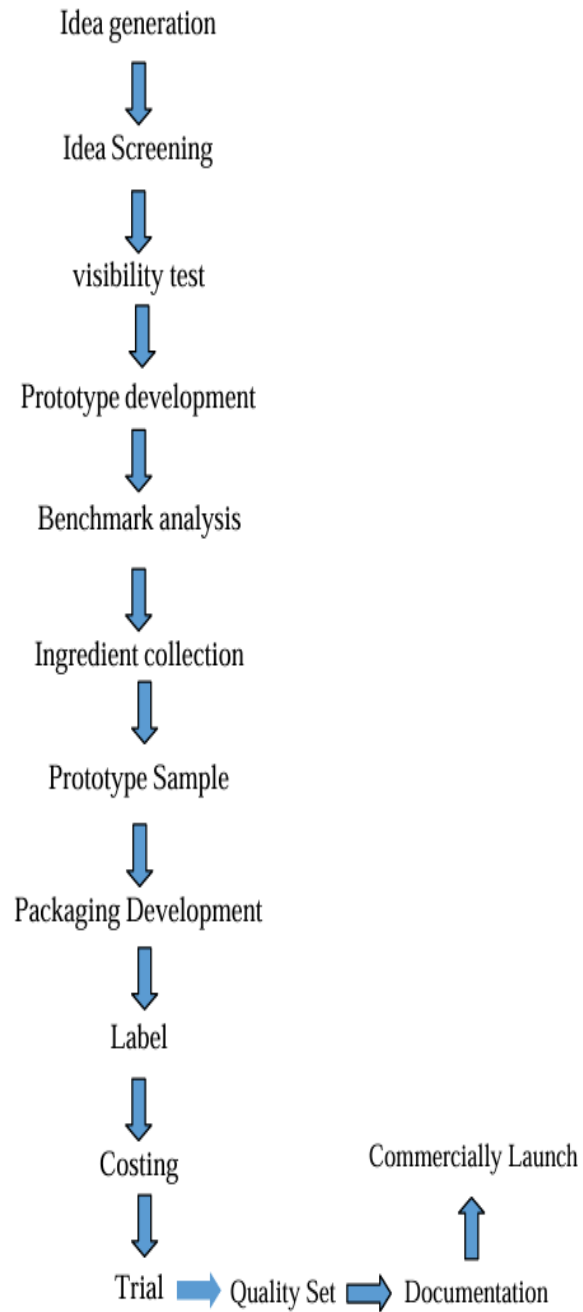


Figure 2.2: R&D Product Development

In every department of AFBL group following this organogram:

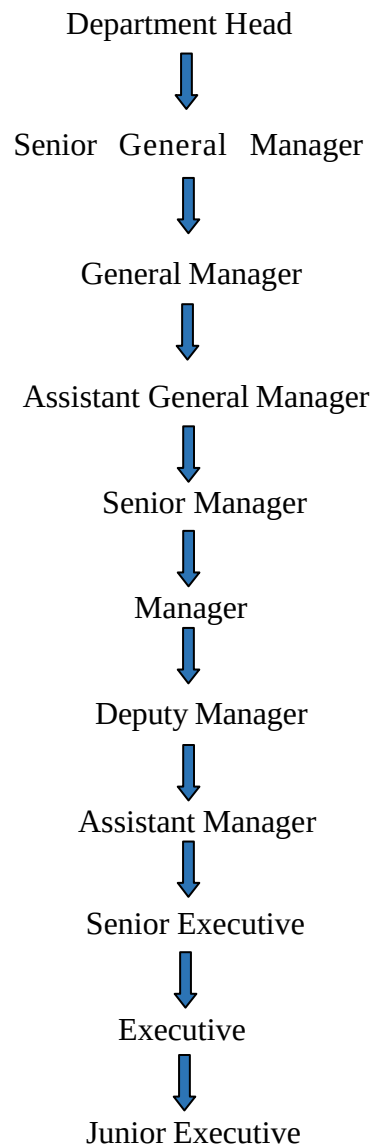


Figure 2.3: Organogram

2.5 Production System of AFBL

The practical expression of AFBL's people and customer-oriented philosophy is known as the AFBL production system. AFBL is not a strict company-imposed procedure but a set of principles that have been proven in day-to-day practice over many years. Many of these ideas have been adopted and imitated all over the world.

AFBL production system has three desired outcomes:

- To provide the customer with the highest quality product at the lowest possible cost and time with the shortest possible lead times.
- To provide members with work satisfaction, job security, and fair treatment.
- It gives the Company flexibility to respond to the market, achieve profit through cost reduction activities and long-term prosperity.

AFBL production system strives for the absolute elimination of waste, overburden, and unevenness in all areas to allow members to work smoothly and efficiently. The foundation of the AFBL production system is built on standardization to ensure a safe method of operation and a consistent approach to quality. AFBL members seek to continually improve their standard processes and procedures to ensure maximum quality, improve efficiency and eliminate waste.

In AFBL they mainly follow this organizational lineup:

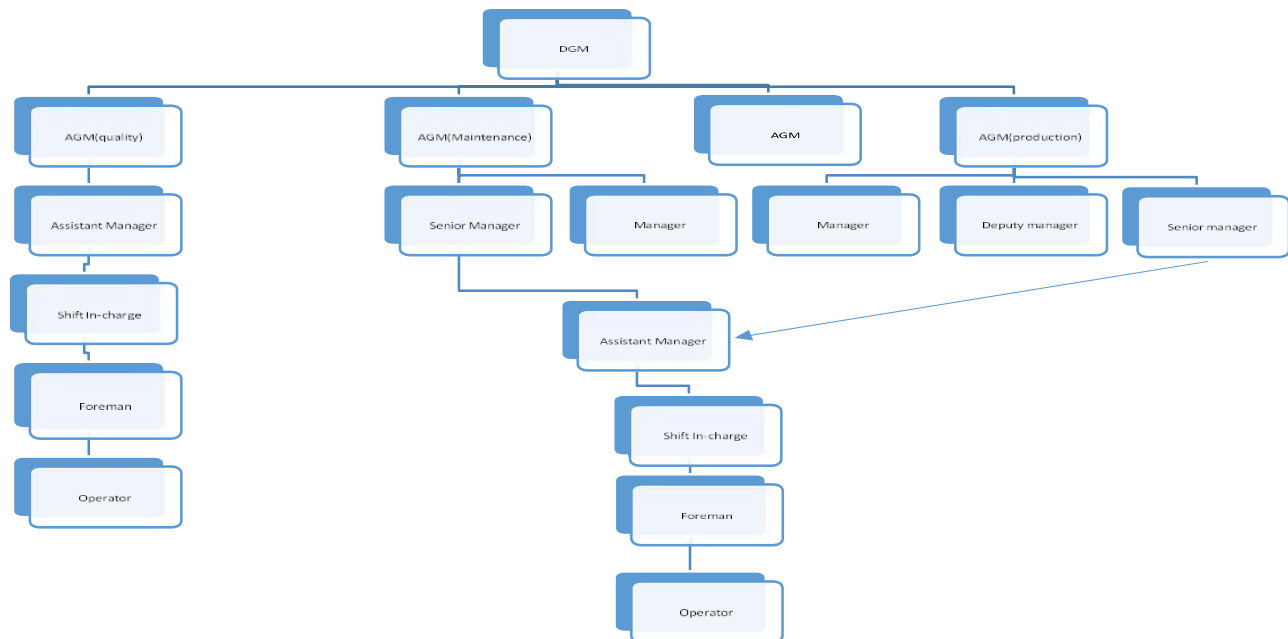


Figure 2.4: Organization Set Up

2.6 Job Responsibilities of Officer

- Take care of commercial papers /documentation for import and export.
- Keep liaise with the concerned department like Customs/Shipping Agent/C&S/Bank etc.
- Prepare necessary statement and report in line with the management's requirement
- To make correspondence with overseas office/external parties/ customers as and when necessary.
- Any other related duties assigned by the management
- Keeping record of all buyers' original documents submitted to bank manually inregister for individually.
- Keeping record of back-to-back L/C performance for all buyers individuallytoavoid excess (%) of back-to-back limit.
- Keeping record of original documents to bank doing reconcile those with ex-factory shipment so that no document missing to bank.
- Bring Bill no., Proceeds and Bank to Bank Courier no. from bank, a day weekly.
- Have sign up job assignment as responsibility.
- Prepare EP, IP & submit to Distribution office.
- Issuing consumption report & maintain bond resister.
- Handle the import/export related activities as far as the commercial is concerned.
- Look after the issues in relation with L/C for import and export.
- Co-ordinate with logistic, CS as well as CNF to ensure smooth delivery
- Prepare consumption statement while C&F come to our premises to take goods.
- Commercial Officer Job Update the delivery status-After completion the delivery.
- To make the Audit report (Export Proceed Realization Certificate) individually.

CHAPTER 3

PRODUCTION SYESTEM

3.1 Description of Production

AFBL food system is all processes and infrastructure involved in satisfying the population's food security: gathering, growing, harvesting, storing, processing, packaging, transporting, marketing, consuming of food, and disposing of food waste.

A food production system has three parts:

- Input: The different ingredients, materials, machinery, and items, which go into the system.
- Process: The various things that happen to the Input which change it into the Output.
- Output: The finished food product.

Table 3.1: Products Description

Product	SKU	Pack size
Mojo	250 ml Can (light)	24
	250 ml Can (Zero)	24
	250 ml Can (Naga)	24
	250 ml Can	24
	250 ml Pet	24
	500 ml Pet	24
Clemon	250 ml Can	24
	250 ml Pet	24
	500 ml Pet	24
Twing	250 ml Pet	24
	500 ml Pet	24
Houston	250 ml Pet	24
Rivera	500 ml Pet (24)	24
	1000 ml Pet	12
	1500ml Pet	6
	2250ml Pet	6

SPEED	250 ml Can	24
Farm Fresh	500ml UHT	16
Ghee	100ml	1
	200ml	1
	400ml	1
	450ml	1
	900ml	1
Chips	Cheese Puffs(20g)	48
Flavored Milk	Farm Fresh Chocolate 250 ml Pet	24
	Farm Fresh Elachi 250 ml Pet	24
	Farm Fresh Mango 250 ml Pet	24
Powder Milk	Farm Fresh Powder Milk-100 gm- Pack	30
	Farm Fresh Powder Milk-200 gm- Pack	24
	Farm Fresh Powder Milk-500 gm- Pack	12
Aafi Puffed Rice	Aafi Puffed Rice-250gm	30
	Aafi Puffed Rice-500gm	16
Afi Chanachur	Aafi Chanachur Classic-35 gm	72
	Aafi Chanachur Classic-150 gm	24
	Aafi Chanachur Classic-300 gm	12
	Aafi Chanachur Jhal-35gm	72
	Aafi Chanachur Jhal-150gm	24
	Aafi Chanachur Jhal-300gm	12
Mojo	1000 ml Pet	12
	2000 ml Pet	6
Lemu	250 ml Can	24
	250 ml Pet	24
Clemon	1000 ml Pet	12
	2000 ml Pet	6
Clear up	250 ml Pet	24

	500ml Pet	24
SPEED	250 ml Pet	24
Houston	250ml Can	24
Wild Brew	250ml Can	24
	250ml	24
	500ml	24
Frutika	125ml(Little Frutika)TPK	48
	125ml LittleFrutika-SIG	48
	250ml (Prizma Mango) TPK	24
Farm Fresh	200ml TPK	30
SPA	250ml	24
	330ml Pet	24
	500ml Pet (24)	24
	500ml Pet (24)	24
	1000ml Pet	12
	1500ml Pet	6
	2250ml Pet	6
	5000ml Pet	2
Chips	Cheese puffs(25g)	48
Chips	O Potatoes	48
Bar	Mango bar(15gm)	300
	Mango bar small(10gm)	300
	Mango barjar(15gm)	300
	Mango bar jarsmall(10gm)	300
Powder Milk	Farm Fresh PowderMilk-50 gm- Pack	60
	Farm Fresh PowderMilk-400 Gm	12
	Farm Fresh PowderMilk-1000 gm	6
Farm Fresh Laban	Farm Fresh Laban250ml	24
Farm Fresh Lassi	Farm Fresh Lassi250ml	24
	Farm Fresh Lassi 200ml	

	Farm Fresh Lassi180	
Flavor Milk	Farm Fresh Chocolate MilkUHT (200ml)	30
	Farm Fresh MangoMilk UHT (200ml)	30
Aafi Toast	Aafi Sweet Toast-200gm	12
	Aafi Sweet Toast-350 gm	6
	Aafi Plain Toast-350 gm	6
Aafi Dry Cake	Aafi Dry Cake-40gm	48
	Aafi Dry Cake-160 gm	12
	Aafi Dry Cake-350 gm	6
Yummy Lassi	Yummy Lassi180ml	
	Yummy Lassi200ml	
	Aafi Dry Cake-160 gm	12
	Aafi Dry Cake-350 gm	6

3.1.1 Brief description of Production System

The production system of an organization is that part, which produces products of an organization. It is that activity whereby resources, flowing within a defined system, are combined and transformed in a controlled manner to add value in accordance with the policies communicated by management. A simplified production system is shown above. The production system has the following characteristics:

3.1.2 Classification of production System

Production systems can be classified as Job Shop, Batch, Mass and Continuous Production systems.

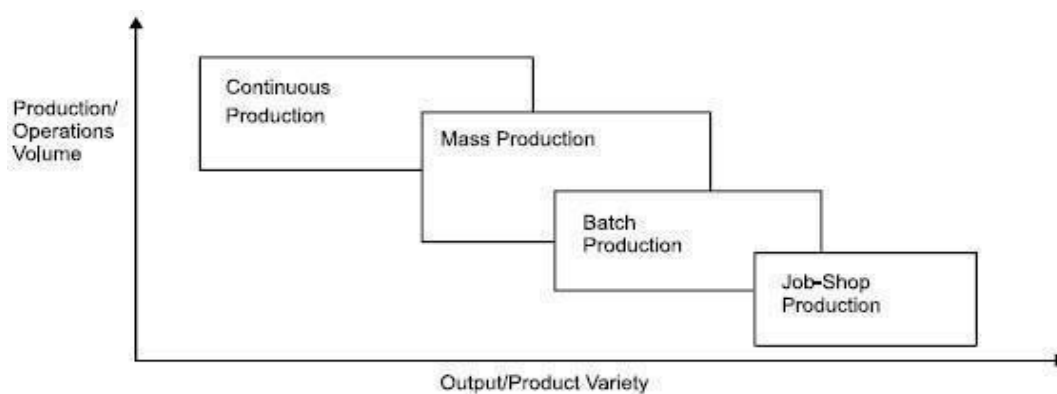


Figure 3.1: Production System Classification

3.1.3 Batch Production

AFBL mainly follow batch production. Batch production is defined by American Production and Inventory Control Society (APICS) “as a form of manufacturing in which the job passes through the functional departments in lots or batches and each lot may have a different routing. It is characterized by the manufacture of limited number of products produced at regular intervals and stocked awaiting sale.

3.1.4 Characteristics

- Batch production system is used under the following circumstances:
- When there are shorter production runs.
- When plant and machinery are flexible.
- When plant and machinery set up is used for the production of item in a batch and change of set up is required for processing the next batch.
- When manufacturing lead time and cost are lower as compared to job order production.

3.1.5 Advantages

- Better utilization of plant and machinery.
- Promotes functional specialization.
- Cost per unit is lower as compared to job order production.
- Lower investment in plant and machinery.
- Flexibility to accommodate and process number of products.

Table 3.2: Machineries List

Types of Machines	Amounts
Blow Mold	8 (approx.)
Filler	11 (approx.)
Check mate	8 (approx.)
Date code m/c	22(approx.)
Label m/c	11(approx.)
Wrapping m/c	11(approx.)
Palletizer m/c	8(approx.)
Handle m/c	4(approx.)
De- palletizer m/c	2(approx.)
Buffer m/c	2(approx.)
Homogenizer m/c	3(approx.)
Woven m/c	6(approx.)
Slicer	3(approx.)
Mixing	4(approx.)
Seamer	2(approx.)
Packaging	3(approx.)
Dryer	1(approx.)
Crusher m/c	2(approx.)
Washing m/c	2(approx.)
Requite m/c	1(approx.)
Angle m/c	1(approx.)
High cap m/c	6(approx.)
Printing m/c	4(approx.)
Film making m/c	3(approx.)
Calleting m/c	2(approx.)

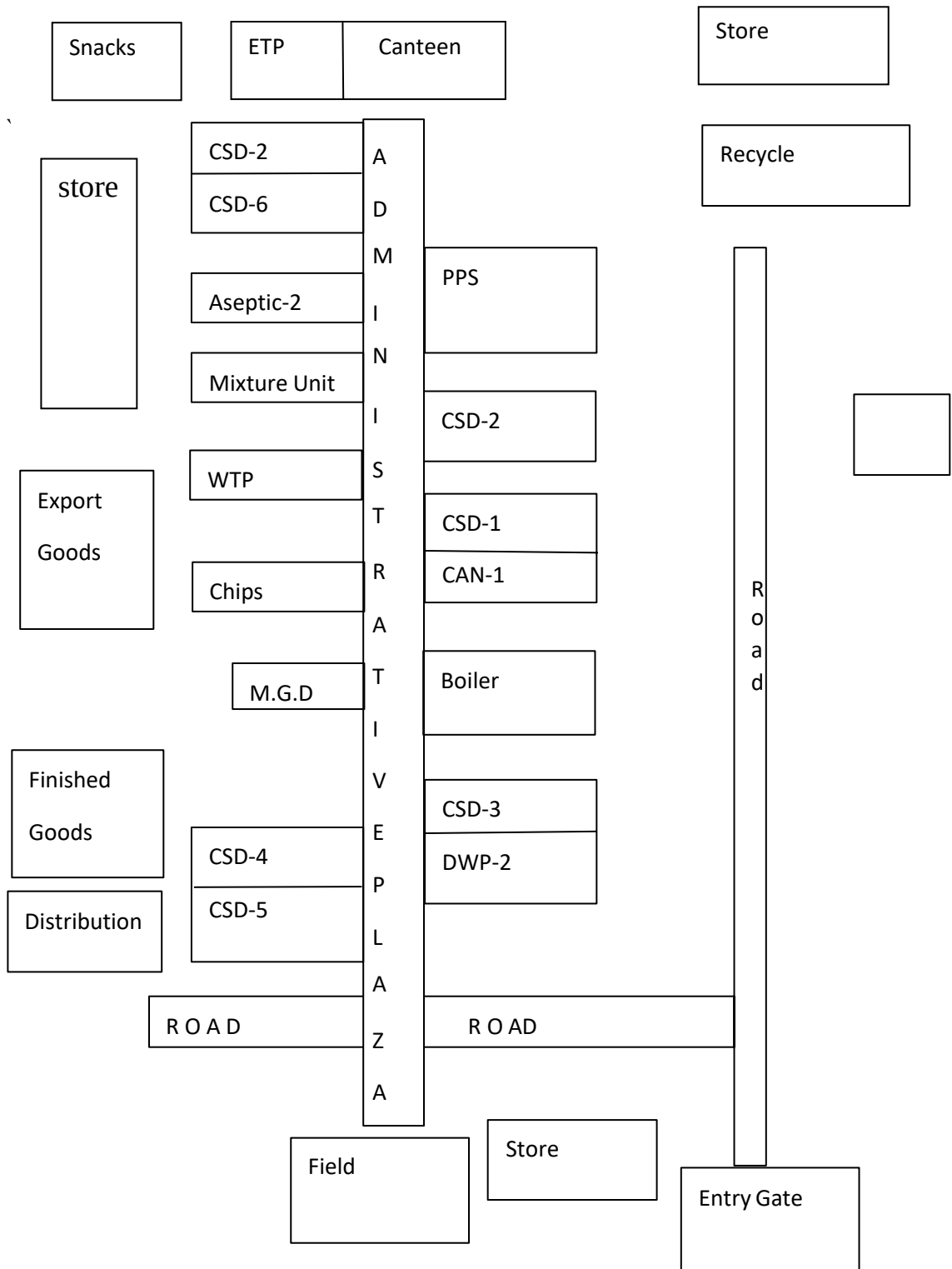


Figure 3.2: Plant Layout

3.1.6 Sugar Processing Plant

The production of sugar syrup is a critical process at Akij Food and Beverage Limited, ensuring that the base ingredient for carbonated soft drinks is of the highest quality. The following detailed description outlines each step of the sugar to syrup production process, emphasizing its importance and the role it plays in the overall production workflow.

1. Sugar Dumping

Received solid sugar is introduced into the processing system. This initial step involves transferring bulk sugar from storage containers into the sugar dissolver. Efficient handling during this phase is crucial to maintain the quality and purity of the sugar.

2. Sugar Dissolver

The solid sugar is dissolved in hot water maintained at a temperature of 40-45°C. The dissolving process takes approximately 5-7 minutes, transforming the solid sugar into a liquid solution. This conversion is essential for creating a homogenous syrup that can be easily processed and filtered in subsequent steps.

3. Pasteurization

The sugar solution undergoes pasteurization by heating it to 80-85°C for 42 seconds. This critical step eliminates potential microorganisms, ensuring the syrup is safe for consumption and extends its shelf life. Pasteurization is fundamental in maintaining the microbiological safety of the syrup.

4. Reaction Tank

Post-pasteurization, the sugar solution is transferred to a reaction tank where it is mixed with activated carbon. This mixture is held for 60 minutes, allowing the activated carbon to adsorb impurities, odors, and colors from the syrup. The reaction tank ensures that the syrup is purified, enhancing its quality and clarity.

5. Filter 1 & 2

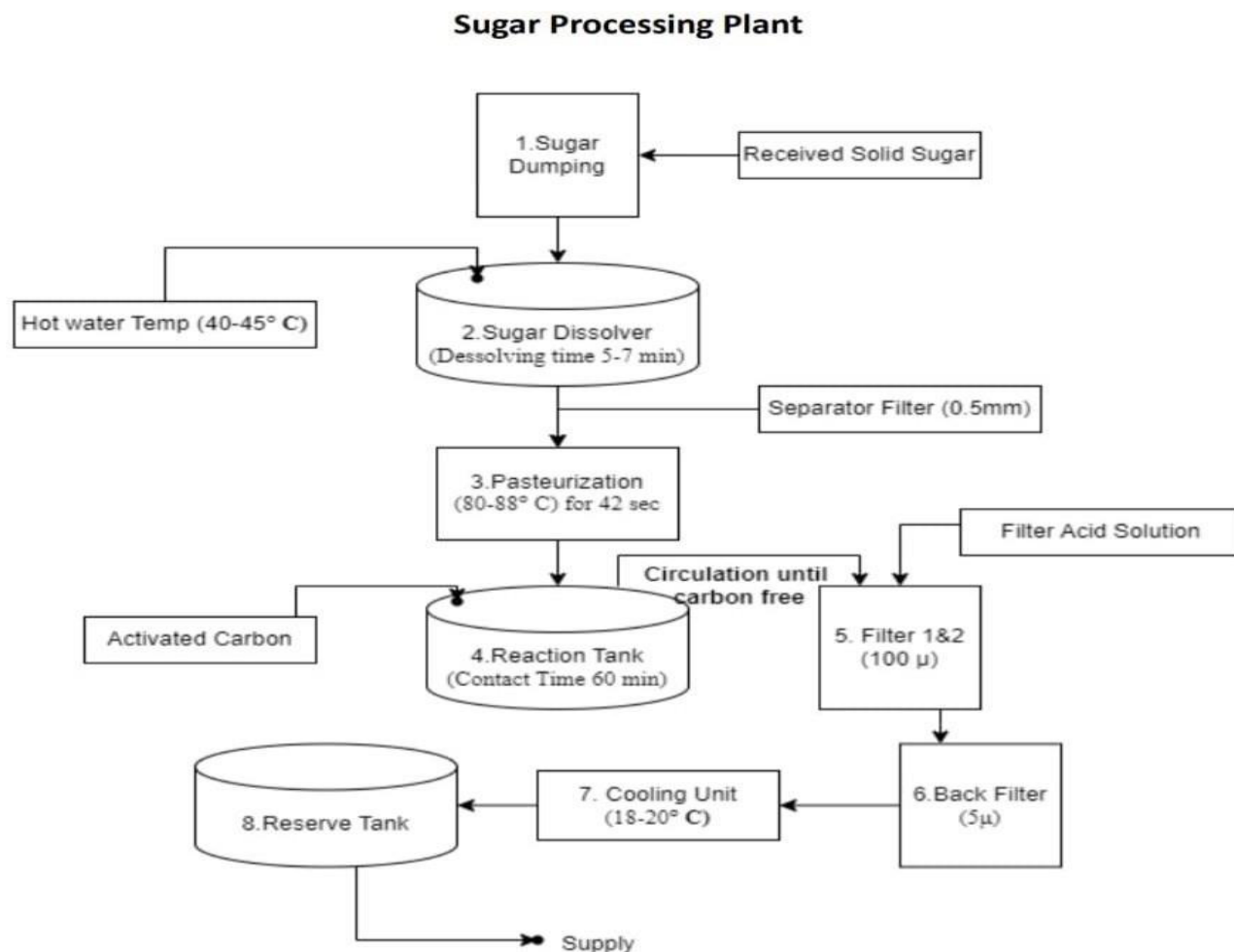
The purified sugar solution is then passed through a series of filters with a pore size of 100 micrometers. These initial filters remove larger particulate impurities that may be present, ensuring the solution is free from coarse contaminants.

6. Back Filter

Further filtration is carried out using a finer filter with a pore size of 5 micrometers. This step is crucial for removing finer particles, resulting in a high-purity sugar solution that meets the stringent quality standards required for beverage production.

7. Cooling Unit

The filtered sugar solution is cooled to a temperature range of 18-20°C. Cooling is a vital step as it prepares the syrup for storage or immediate use in production, preventing microbial growth and maintaining the integrity of the syrup.



3.1.9 Products in CSD-3



Speed-pet 250 ml



Twing-pet 250 ml



Mojo-pet 250 ml

Figure 3.3: Products in CSD-3

3.1.10 Ingredients of products in CSD(3) Speed (pet)

Carbonated Water, refined Sugar, Critic Acid (Acidulate E-330, Tri-Sodium Citrate (Acidity regulator E-331), Sodium Benzoate (preservative E-211), Caffeine, D-Alpha-Tocopherol (Vitamin –E), permitted Food Color (E-102 & E-110) & Flavoring.

Twing (pet)

Treated Water, Refined Sugar, Carbon Dioxide, Sodium Benzoate (Preservative - E-211). Tri-Sodium Citrate (Acidity Regulator, E-331), Sodium Chloride, Citric Acid (Acidulate, E- 330)& lemon Flavor.

Sizes: 250ml, 500 ml, 1 liter

Mojo (pet)

Carbonated Water, Refined Sugar, Carbon Dioxide, phosphoric acid (acid lent - E-338). Caramel color(E-150d), caffeine & Flavoring.

Sizes: 150ml can, 250ml pet and can, 500ml pet, 1-liter pet, 2-liter pet.

3.1.11 Process flow chart of CSD-3

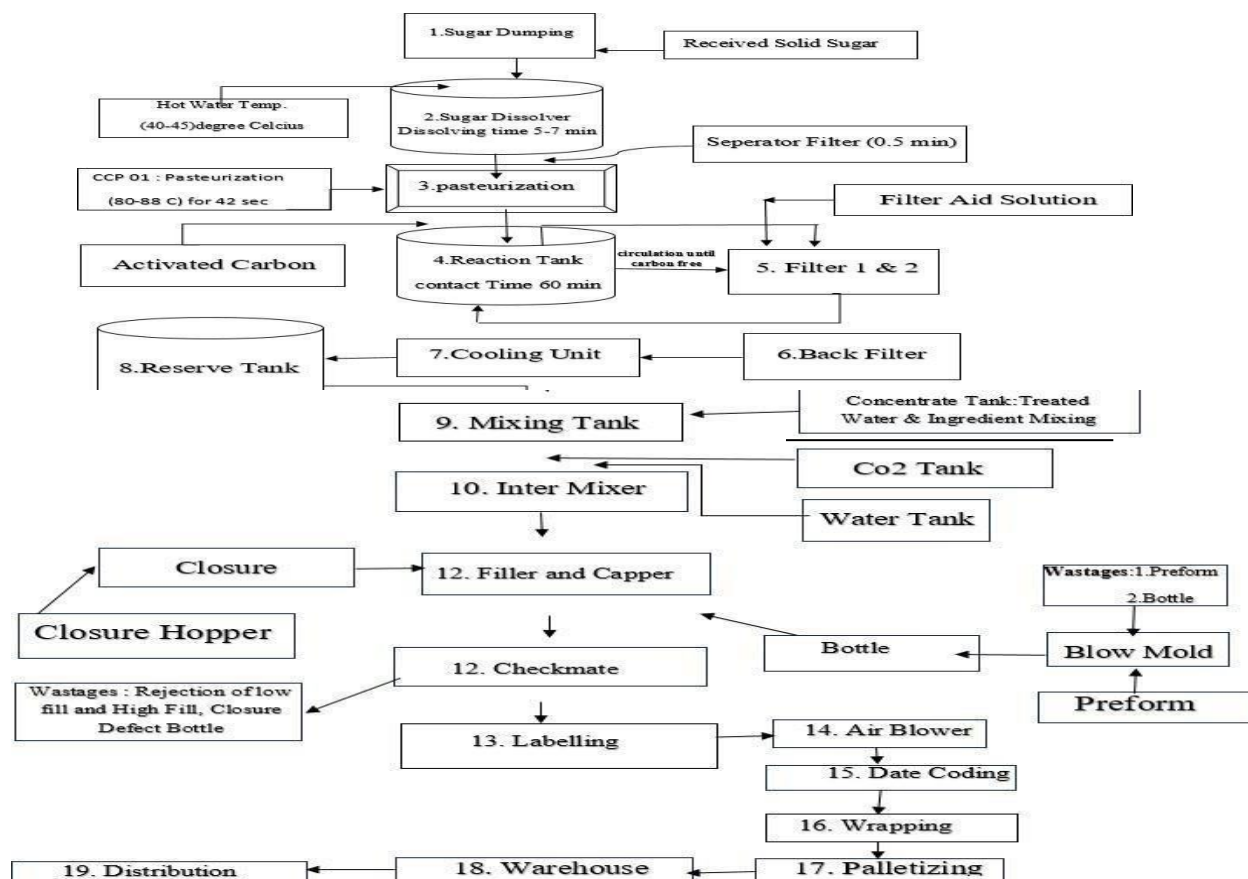


Figure 3.4: Process flow chart of CSD-3

3.2 Brief description of production line (CSD-3)

Production line where mainly pet carbonated soft drinks are produced. Man power of this production line 18/shift. Total no. of machine is 11. Capacity of production 45,000/hour pet bottles. Set temperature for preform is 108° C. Cooling temperature is 8 – 12°C. Pre-blowing pressure is 11.5 bar & find blowing pressure 28.5 bar. Wrapping temp. is 210°C.

3.3 Brief description of production line (Can-1)

Production line where mainly Speed-Can, Carbonated soft drinks are produced. Man power of this production line 19/shift. Total no. of machine is 9. Capacity of production 18,000/hour can, 750 case/hour. Cooling temp. 16.9° C. Wrapping temp. is 210°C. CIP is take over in every 72 hour.

3.3.1 Products of (can-1)



Figure 3.5: Products of Can-1

3.3.2 Ingredients of products in Speed Can-1

Carbonated Water, refined Sugar, Critic Acid (Acidulate E-330, Tri-Sodium Citrate (Acidity regulator E-331), Sodium Benzoate (preservative E-211), Caffeine, D-Alpha-Tocopherol (Vitamin –E), permitted Food Color (E-102 & E-110) & Flavoring.

Clemon-Can

Carbonated Water, Refined Sugar, Citric Acid(E-330), Tri-Sodium Citrate (E-331) Sodium benzoate (E-211). Sodium Chloride & permitted Artificial Flavor.

Mojo (Can)

Carbonated Water, Refined Sugar, Carbon Dioxide, phosphoric acid (acid lent-E-338). Caramel color (E-150d), caffeine & Flavoring.

3.3.3 Production Flow Chart of (Can-1)

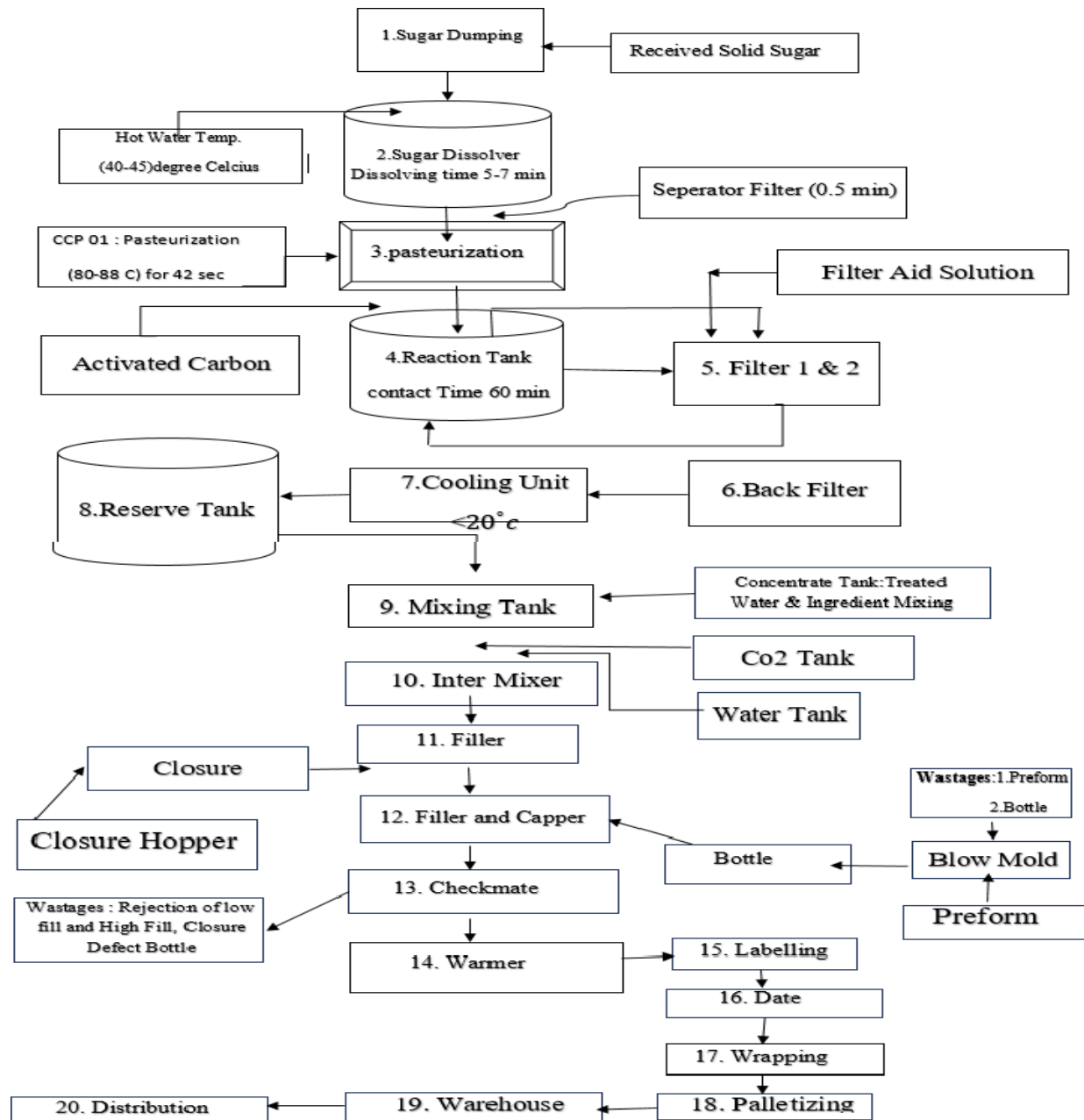


Figure 3.6: Production Flow Chart of Can-1

3.4 Brief description of production line (CSD-1)

Production line where mainly pet carbonated soft drinks are produced. Man power of this production line 19/shift. Total no. of machine is 9. Capacity of production 18,000/hour pet bottles. Wrapping temp. is 210°C. Closure torque 18 LB & range 1b.

3.4.1 Products of (CSD-1)



Figure 3.7: Products of CSD-1

3.4.2 Ingredients of products in (CSD-1)

Houston: Carbonated water, refined sugar, citric acid (Acidulant E-330), Tri-sodium citrate (Acidity regulator E-331), Sodium benzoate (preservative E-221), Caffeine, Food color (E-102 & E-133) & Flavoring.

Mojo (pet): Carbonated Water, Refined Sugar, Carbon Dioxide, Phosphoric Acid (Acidulant - E-338), Caramel color (E-150d), caffeine & Flavoring.

3.4.3 Production Flow Chart of (CSD-1)

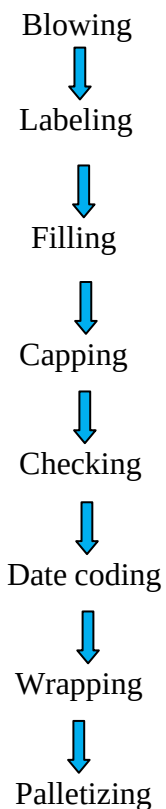


Figure 3.8: Production Flow Chart of CSD-1

3.5 Brief description of production line (CSD-4/5)

Production line where mainly Speed pet, Mojo pet, Clemon pet & Twing Carbonated soft drinks are produced. Man power of this production line 15/shift. Total no. of machine is 11. Capacity of production 72,000/hour pet. Base Cooling temp. 2° C. Wrapping temp. is 210°C. B/M 95°C & O.V.T 42°C. 1 label case can used for 10-12 thousand pet bottles. Wrapping speed 60 case/min. Palletizer machine capacity 66240 pcs/ hour.

3.5.1 Products of CSD (4/5)



Figure 3.9: Products of CSD-4,5

3.5.2 Ingredients of Products in CSD (4/5)

Clemon (pet):

Carbonated Water, Refined Sugar, Citric Acid(E-330), Tri-Sodium Citrate (E-331) Sodium benzoate (E-211). Sodium Chloride & permitted Artificial Flavor.

Mojo (pet):

Carbonated Water, Refined Sugar, Carbon Dioxide, phosphoric acid (acid lent - E-338). Caramel color(E-150d), caffeine & Flavoring.

3.5.3 Production flow Chart of CSD (4/5)

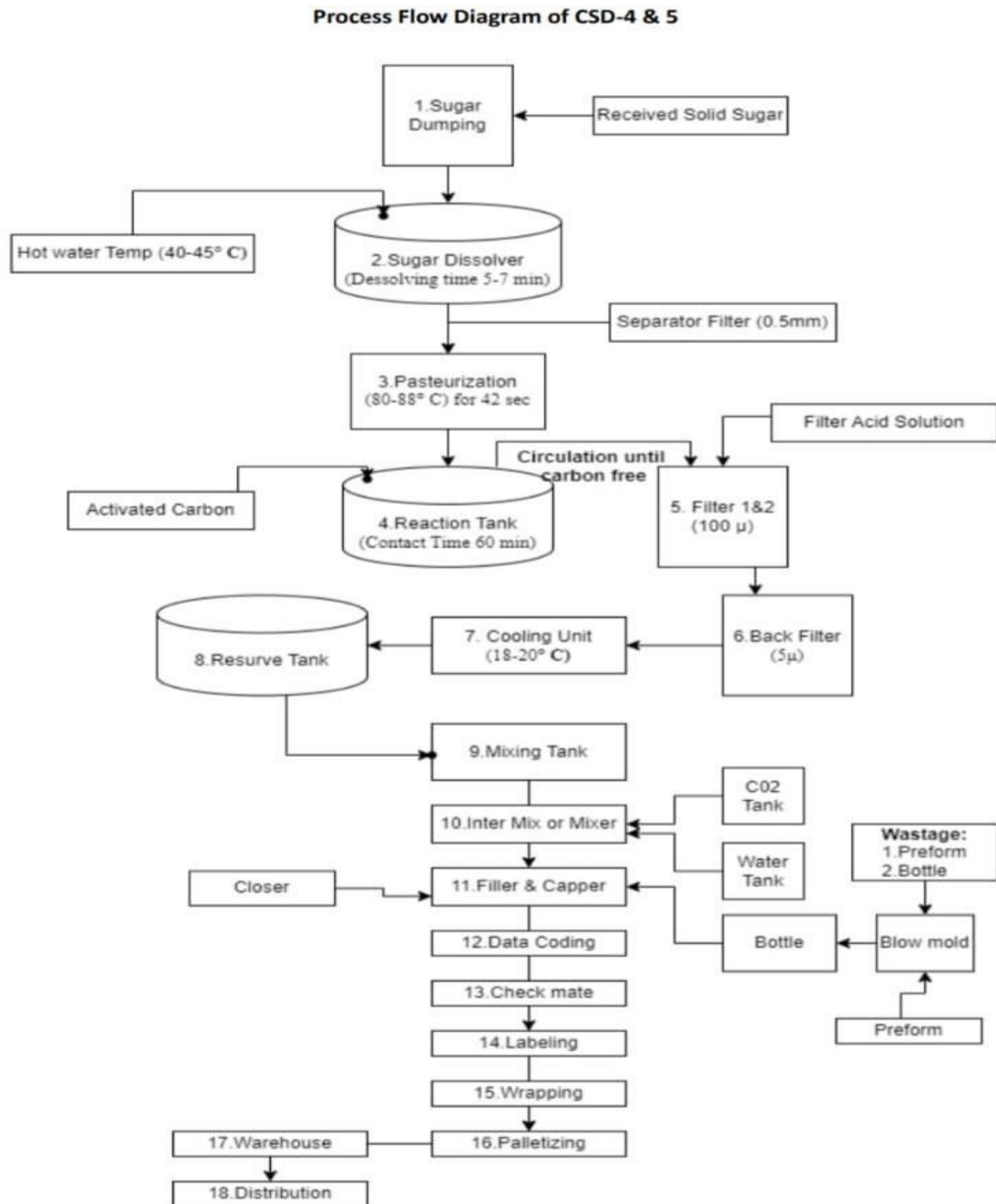


Figure 3.10: Production flow Chart of CSD 4,5

3.6 Brief description of production line (New- Aseptic)

Production line where mainly pet aseptic fruit drinks are produced. Man power of this production line 18/shift. Total no. of machine is 8(approx.). Machine Capacity 54,000/hour but filling capacity 32,750/hour. Set temperature for blow mold is 330° C. Filling Temperature is 30 – 32°C. Cooling. Pre-blowing pressure is 15.5 bar. Wrapping temp. is 210°C. Filling Value of Bottle is 32-48. For labeling air pressure temperature is used. CIP is taken over in every 72 hours. There is no preservative is used in aseptic food drinks.

3.6.1 Products of New-Aseptic



Figure 3.11: Products of New Aseptic

3.6.2 Ingredients of Products (New Aseptic)

Frutika:

Water, Mango Pulp, Refined Sugar Stabilizer, Xanthan Gum (E-330), Tri- Sodium Citrate (E-331), Permitted Nature Identical Food Grade Color (E-160a) & Permitted Natural Flavor.

Aafi:

Water, Refined Sugar, Mango Pulp, Sodium Benzoate (E-211), Stabilizer: Xanthan Gum (E-415), Ascorbic Acid (E-300), Citric Acid (E-330), Tri-Sodium Citrate (E-331), Permitted Nature Identical Food Grade Color (E-160a) & Permitted Natural Flavor & Aseptically Filled.

Farm Fresh Chocolate Flavor:

Pure Cow's Milk, Cocoa Powder, Refined Sugar, Stabilizer, skimmed milk powder, Treated Water.

Aafi Litchi:

Water, refined sugar, citric acid (acidulate E-330), Stabilizer, Car boxy methyl cellulose (E-4660, malic Acid (E-466), Malic Acid (E-2960, Tri sodium Citrate (Acidity regulator E- 331), Cloudifier & Flavoring.

Drink Eat:

Water, refined sugar, stabilizer, carboxyl methyl cellulose (E-466), glen gum (E-418), citric acid (E-330), Ascorbic Acid (E-300), Sodium lacked, Neutral clued, sodium benzo-yet (E- 211), potassium Sorbet (E-202) & Added flavor.

Production Layout of New-Aseptic

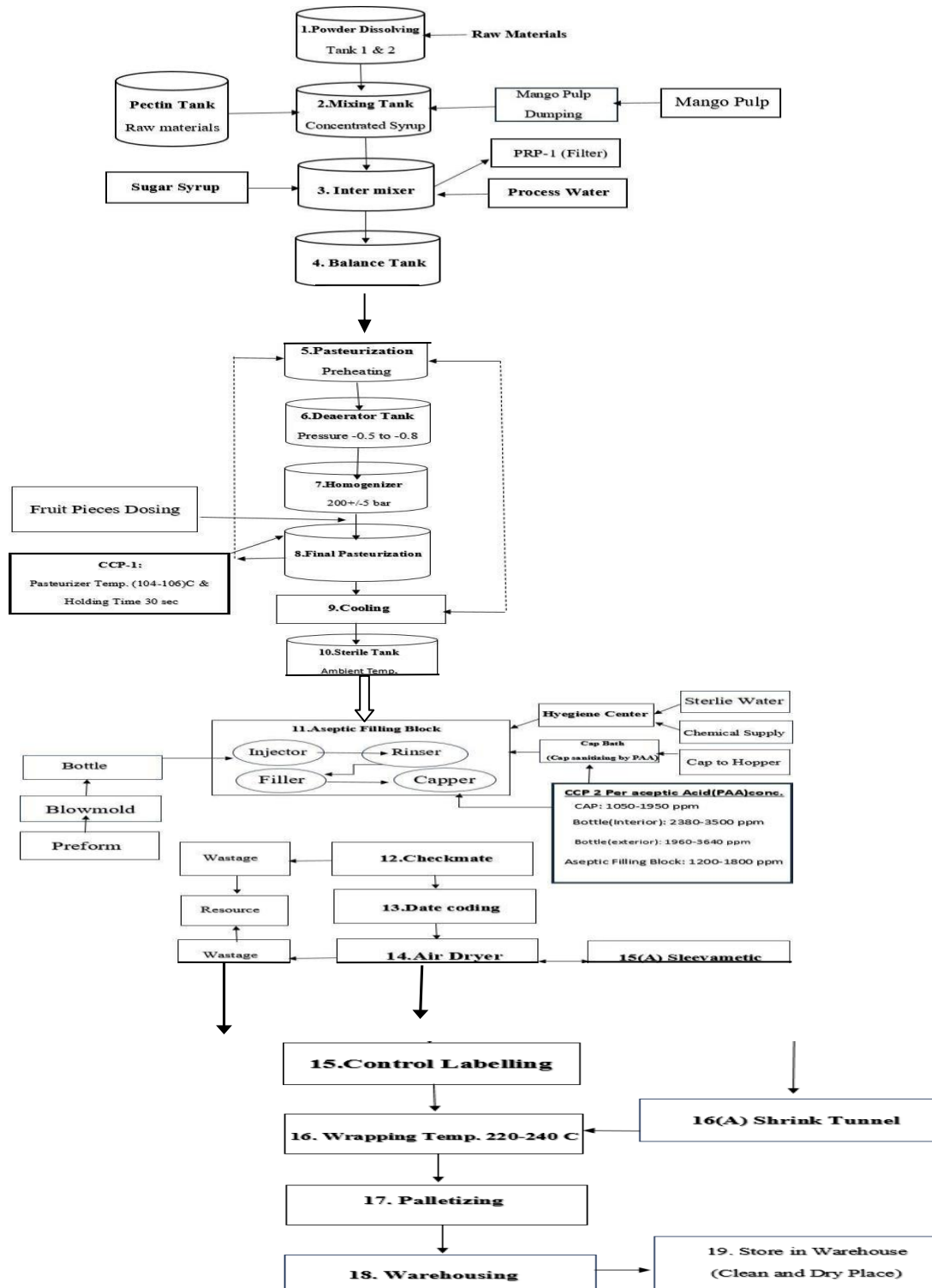


Figure 3.12: Production Layout of New Aseptic

3.7 Brief description of production line (CSD-6)

Production line where mainly pet speed carbonated soft drinks are produced. Man power of this production line 14/shift. Total no. of machine is 10 (approx.). Machine Capacity 81,000/hour but maximum filling capacity 75,000-78,000/hour. Set temperature for blow mold is 160° C. Oven Temperature is 59°C. Cooling Temperature is 2 – 15°C. Pre-blowing temperature is 15°C. Filling pressure is 4:8 bar. Wrapping Temperature is 210°C. Filling Value of Bottle is 32-48. CIP is taken over in every 108 hours. 192 Bottles are filled at a time. Capper head is 48, torque 19 LBS ft. & Capping Pressure is 220 N. Chemical Spray is taken over in every 54 hours. Checkmates used to check under-fill, overfill, fault in cap.

3.7.1 Products of CSD-6

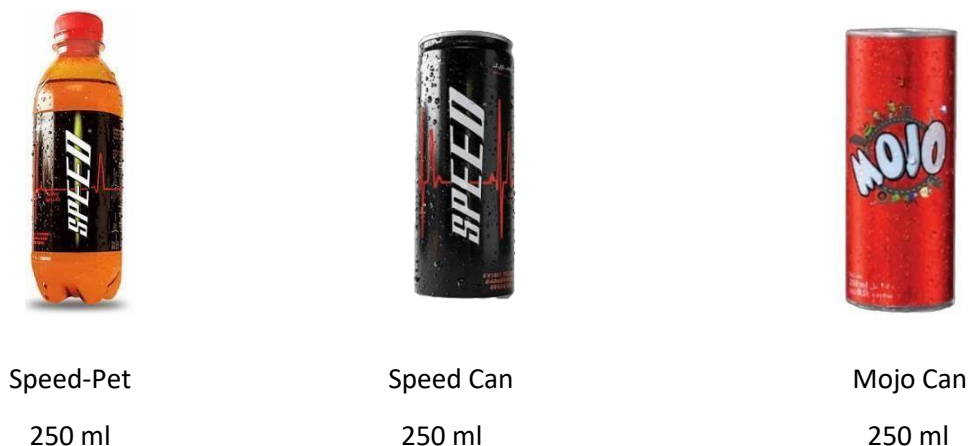


Figure 3.13: Products of CSD-6

3.7.2 Ingredients of Products (CSD-6)

Speed (pet)

Carbonated Water, refined Sugar, Citric Acid (Acidulate E-330, Tri-Sodium Citrate (Acidity regulator E-331), Sodium Benzoate (preservative E-211), Caffeine, D-Alpha-Tocopherol (Vitamin –E), permitted Food Color (E-102 & E-110) & Flavoring.

Mojo (pet)

Carbonated Water, Refined Sugar, Carbon Dioxide, phosphoric acid (acidulant - E-338). Caramel color(E-150d), caffeine & Flavoring.

3.7.3 Production Layout of (CSD-6)

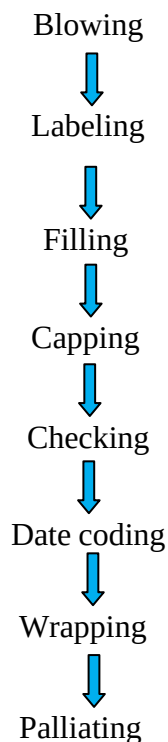


Figure 3.14: CSD-6 Production Layout

3.8 Brief description of production line (Can-2)

Production line where mainly Non-Alcoholic Malt Drinks are produced. Man power of this production line 13/shift. Total no. of machine is 9. Machine Capacity 36,000/hours (250 ml can) & 18,000/hours (300ml can). Blowing temperature is 8 – 12° C. warming Temperature is 30 – 35°C. Cooling Temperature is 2 – 15°c. Pre-blowing temperature is 15°c. Wrapping Temperature is 210°C. CIP is taken over in every 96 hours. There is 32 filling bulbs, 6 lid in seamer machine in this production line. It contains 24 pieces of can in 1 case. Empty can mainly have imported from Sri Lanka, Pakistan & UAE.

3.8.1 Ingredients of product in (Can-2)

Wild-brew:

Carbonated water, Refined sugar, malt compound, citric acid (E-330), Ascorbic acid (E-300), Sodium- benzo yet (E-221) & Flavoring.

3.8.2 Production layout of (Can-2)

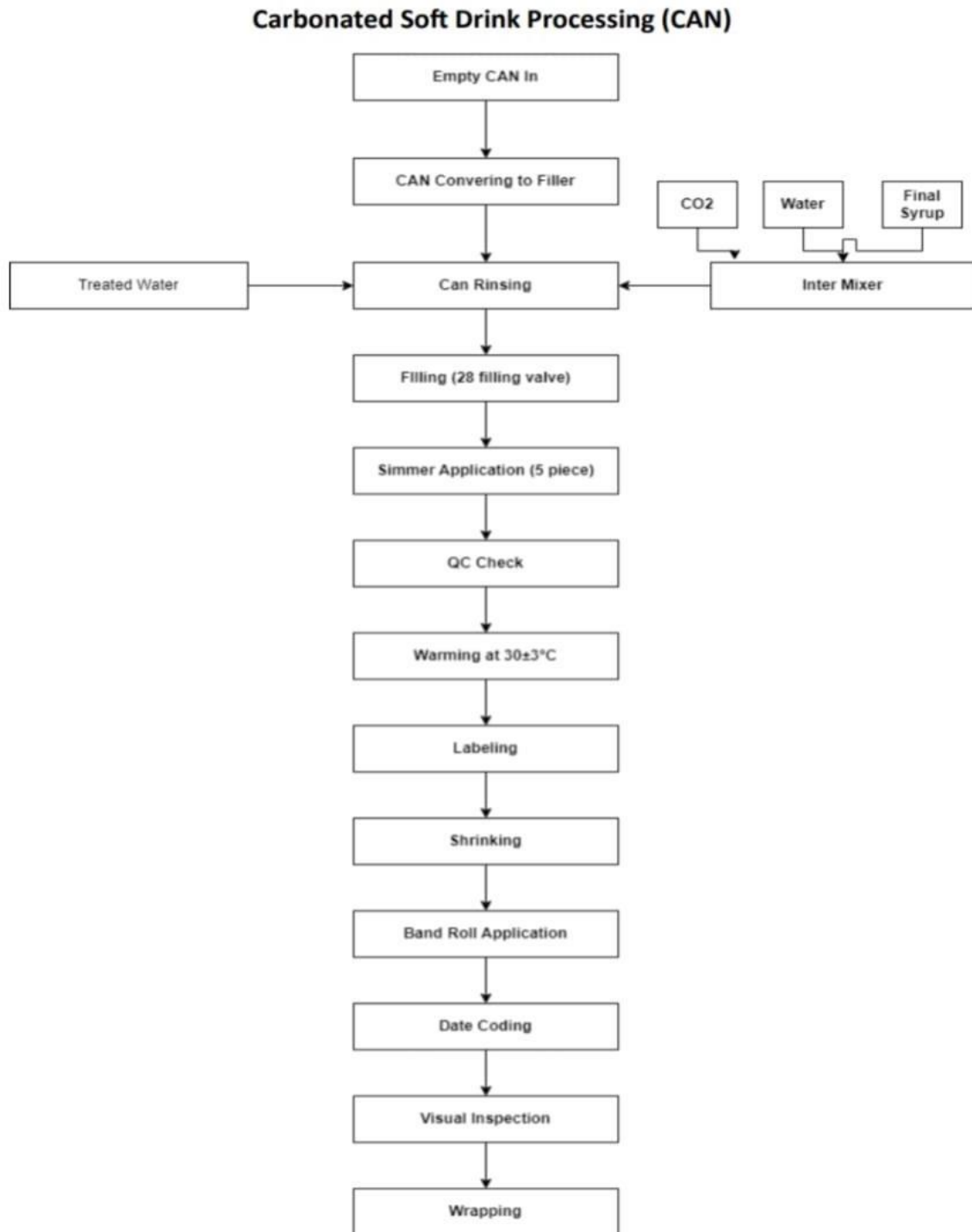


Figure 3.15: Can-2 Production Layout

3.9 Brief description of production line (Daily-Snacks)

Production line where Daily food items are produced. Man power of this production line 40/shift for snacks-1, 46/shift for bakery, 12/shift for puffed rice production, 9/shift for chips production line. Total no. of machine is 24(approx.). Machine Capacity 44 pack/min. for chips, 400-500 kg/hour for puffed rice, 420 kg rolling/ hour for bakery. In snacks-1(cooling temp is 12 – 14°C, baking temp. 160°C, cooling time 3-4 hours). In puffed rice (mixing time 25-30 min, dryer temp. 160 – 180°C, roster temp. 180 – 220°C.

3.9.1 Products of (Daily Snacks)



Aafi-Puffed Rice
500gm



Aafi Hot Chanachur
1000gm



Cheese Puffs
18gm



Aafi chanachur
25gm, 50gm, 250 gm



Aafi- BBQ Chanachur
25gm, 50gm, 250 gm



O' Potatoes
18 gm

Figure 3.16: Daily Snacks Products

3.9.2 Ingredients of products (Daily Snacks)

Cheese puffs:

Corn Grits, Refined Vegetable Oil, Wheat Flour, Corn Starch, Talcum Powder (Talc; E- 553b), Mixed Seasonings (Cheese Solid, Milk Solid, Refined Sugar, Glucose, Salt Modified Starch (E- 1401), Monosodium Glutamate (E-621), Aspartame (E-962), Permitted Nature Food Grade Color (E-102 & E-110) And Permitted Artificial Flavor.

O Potatoes:

Potato Starch, Potato Flakes, Potato Granules, Edible Vegetable Oil, Salt, Turmeric Powder, Spices & Condiments (Chili, Bit Salt, Sugar Powder, Citric Acid & Seasoning).

Aafi puffed Rice:

Rice & Salt

Aafi Hot Chanachur:

Vegetable Oil, Rock Salt, Lentil, Black paper, Clove, Soybean seasoning, sodium Bi carbonate (E-500), Turmeric powder, Starch, Red Chili Powder, split chickpeas, peanuts, citric acid (E-330), Green beans.

3.10 Brief description of production line (DPW-1)

Production Line Where Mainly Drinking Water Pet Bottle Are Produced. Man Power of This Production Line 10/Shift. Total No. of Machine Is 8(Approx.). Machine Capacity 44,00 Piece/Hour. Preform Weight 44 Gm. 1 Label/5500-Piece Bottle. Per Shift 100 Piece Defect (Approx.). Wrapping Temperature 210°C.

3.10.1 Products of DWP-1



Rivera-Drinking Water

500ml,1000 ml



Spa-Drinking Water

500ml,1000 ml

Figure 3.17: Products of DWP-1

3.10.2 Ingredients of products (DPW-1)

Rivera- Drinking Water: Treated Water

Spa- Drinking Water: Treated Water

3.11 Brief description of Akij Dairy Plant

Production line where dairy milk products are produced. Man power of this production line 18/shift. Total no. of machine is 38(approx.). Machine Capacity 7,300/hour large UHT pack & 9600/hour for small UHT pack. There is 5-line dairy plant. CIP is taken over in every 10 hours. UHT sterilizer is given with high temperature about $137 - 140^{\circ}\text{C}$. Homogenizer is taken over in every 4 seconds about temp. 140°C with 190 Bar Air pressure. Pasteurization is given about 100°C in every 30 seconds with 180 Bar Air pressure.

3.11.1 Products of Akij Dairy Plant



Farm Fresh
Mango Flavored



Farm Fresh
Chocolate Flavored



Farm Fresh
UHT-Milk

Figure 3.18: Dairy Products

3.11.2 Ingredients of Akij Dairy Plant Products:

Farm fresh chocolate flavored:

Pure Cow's Milk, Cocoa Powder, Refined Sugar, Stabilizer, skimmed milk powder, Treated Water.

3.11.3 Production layout of Akij Dairy Plant:

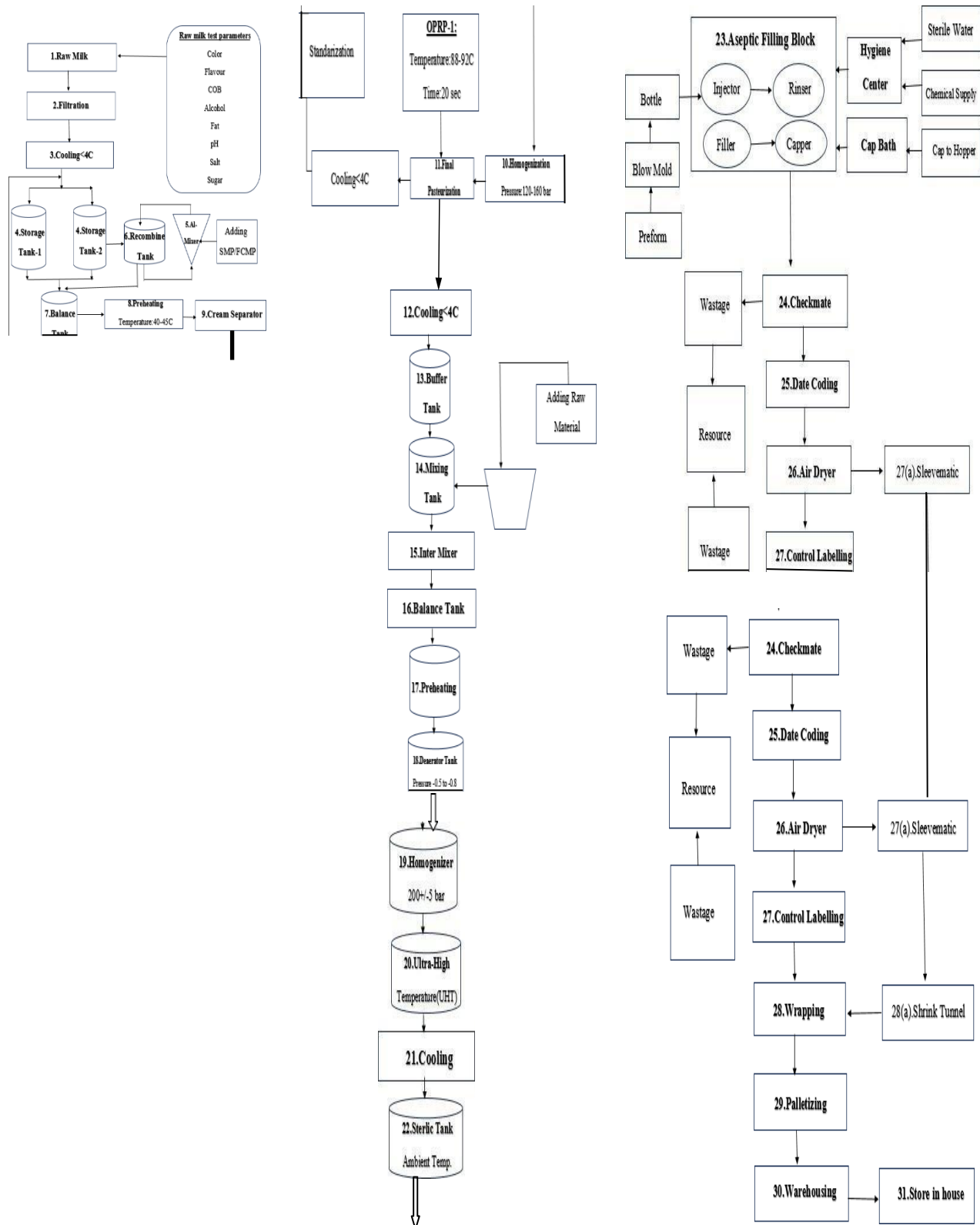


Figure 3.19: Akij Dairy Plant Production Layout

CHAPTER 4

MATERIAL MANAGEMENT

4.1 Types of Material in Storage

A store is a place where excess materials are kept, which will be used as and when required. Loss of items, deterioration, obsolescence, and inadequacy are treated as 'part of life'. The store operation is to receive materials to protect them while in storage from damage and unauthorized removal, issue the material in the right place, and provide these services promptly and at least cost. Storage methods depend on the physical characteristics (size, weight, shape) of the product and the volumes to be stored and handled.

Most commonly used material storage systems:

- Block storage
- Racking system
- Containers
- Shelving system

4.2 Types of Materials

Raw materials for soft drinks & dairy product:

- Carbon dioxide
- Beverage
- Syrup
- Milk

Raw materials for Chips:

- Corn
- White flour
- Food grade telco power
- RBD pump oil

Raw materials for Preform & closure:

- Resin

4.2.1 Packaging Materials:

- Plastics.

4.2.2 Types of Resin

Resins are viscous, liquid polymers derived from organic or synthetic sources. Their benefit centers on their ability to transition from a liquid into a customizable solid, homogeneous structure. In polymer chemistry and materials science, resin is a solid or highly viscous substance of plant or synthetic origin that is typically convertible into polymers. The resin protects the plant from insects and pathogens. Many of these naturally occurring resins have practical uses today and equivalents have been manufactured from petrochemicals. Resin usage as a single, homogeneous form is commonly used in the form of molds as well protective linings and coatings. Classic examples of synthetic polymers include resins and plastics. There are 2 main types of resin: thermosetting and thermoplastic resin.

Table 4.1: Resin Types

Types	Origin
LG Resin	Korea
Gade Resin	India
S-pet Resin	Korea
Gp2 Resin	BD
YS-C01 Resin	China
INO+ Resin	China
10-HR Resin	Thailand

4.3 Procurement System

A procurement system or purchasing system allows for the organizations to automate the process of purchasing of goods or service and maintaining inventory. It helps manage all the procurement related process. It is the core activity of the manufacturing industry. It helps in procurement planning, supplier selection, purchase order creation, invoice verification, and payment management. A typical procurement process includes:

1. Identify a need for products
2. Find suppliers
3. Request for quotation
4. Negotiate with supplies
5. Agree on terms and contracts with supplies
6. Purchase order

4.3 Source

When the raw materials are not found the local market, then go to the foreign market for sourcing raw materials or machine by an ultimate supplier who is ready to supply the product. AFBL focuses on Thailand, China, France, Qatar, Taiwan, Saudi, UAE, Korea, Denmark & Malaysia purchase the raw materials or machine.

4.4 Annual requirements

Annual Requirements & Raw materials of the carbonated soft drinks, Chips & PPS production line:

Table 4.2: Annual Requirements

Raw Materials	Annual Requirement (per year)
Carbon di-oxide	1000 ton (approx.)
Syrup	1125 ton (approx.)
Beverage	3375 ton (approx.)
Corn	1400 ton (approx.)
Wheat Flower	200 ton (approx.)
Telkom Powder	8 ton (approx.)
Starch	50 ton (approx.)
RBD Pump oil	415 ton (approx.)
Poly Bag	250 ton (approx.)
Foil	650 ton (approx.)

4.5 Conclusion

Raw materials and accessories are the major term of any production. We can say that this term is the heart of a AFBL industry. We have tried our label best to collect all data about raw materials though it was not an easy work to do. Moreover, AFBL maintain a good fame by using brand item and qualified raw material.

CHAPTER 5

MANAGEMENT SYSTEM

5.1 Operation Management

Operation management system of AFBL refers to the administration of business practices to create the highest level of efficiency. It is concerned with converting materials and labor into goods and services as efficiently as possible to maximize the profit. Operations management of AFBL attempt to balance costs with revenue to achieve the highest net operating profit possible.

5.1.1 Operation Logistics

Operation Logistics works with factory products and import products. Their primary function is the distribution of the finished goods as per requirement from the production floor. Their job is to receive the product, distributed, and sell them.

5.1.2 Inventory Management System

As for FMC products, it's not allowed to keep finished goods inventory as they don't have any finished goods inventory. They also maintain norms of the products, especially pharma products, all VAT-related functions, and regular stock counting and updating in the **ERP** software. A supply head controls the total logistic system from the head office which is directly connected with the factory's logistic department. Then, the logistics are distributed in the three depots as per the forecasting they got by marketing through sales. Generally, they got 18-month forecasting for distribution.

5.1.3 List of Inventory

- **Raw materials:**
Basic inputs that are converted into finished product through the manufacturing.
- **Working progress:**
Semi-manufactured products need some more works before they become finished goods for sale.
- **Finished goods:**
Completely manufactured products ready for sale.
- **Supplies:**
Office and plant cleaning materials not directly enter production but are necessary for production process and do not involve significant investment.

5.1.4 Scheduling and Sequencing System

At first, the finished goods which need to be distributed come to the quality lab. After they passed the quality check, they are ready for distribution. It takes around one and a half-day to get the quality pass to a specific lot for distribution. Then the logistics received the finished goods and gathered them in the depots. Then they distributed finished goods as per the demand of several distributors from several places, which comes through email. Their job is to deliver the finished goods to the distributor point (DP) with their costing and transport. In this case, they often do clubbing. Clubbing means stock leaving to close locations together. This reduces their cost a lot.

5.2 Quality management

Quality management ensures that an organization, product or service is consistent. It has four main components: quality planning, quality assurance, quality control, and quality improvement. Quality management is focused not only on product and service quality but also on the means to achieve it. Quality management, therefore, uses quality assurance and control of processes and products to achieve more consistent quality. What a customer wants and is willing to pay for it determines quality. It is a written or verbal commitment to a known or unknown consumer in the market. Thus, quality can be defined as fitness for intended use or, in other words, how well the product performs its intended function. There are two main approaches to quality management QC (quality control) and QA (quality assurance). Quality assurance (QA) and quality control (QC) are two terms that are often used interchangeably. Although similar, there are distinct differences between the two concepts. We will explain the differences between quality control and quality management and provide definitions and examples of each.

5.2.1 Quality Control

The process of inspecting products to ensure that they meet the required quality standards.

- Base on inspection
- Takes defects out

5.2.2 Quality Control Test in AFBL

In AFBL there are mainly three main categories of (QC) tests are doing here such as:

- Physical test: bottle, cartoon, length, weight, height & flavor taste.
- Chemical test: titration & formula taste.
- Instrumental test: PH, TDS, Moisture, drying, Ash & Fat

5.2.3 Quality Control Tester Machine in AFBL:

- | | |
|----------------------|----------------------------|
| • Turbidity meter | • Spectro photo-meter |
| • Salt Meter | • Viscometer |
| • TPM | • Auto shaker |
| • Enamel rater | • Melting point machine |
| • Vortex Mixture | • Dryer |
| • Centrifuge | • Bursting strength tester |
| • Conductivity meter | • Desiccator |
| • Jar tester | • Fame hood |
| • PH Meter | • GCMS |
| • Vacuum pump | • ICPE |

5.2.4 Quality Assurance

The processes that ensure product quality meets the requirement of customers. Quality Assurance(QA) is the systematic monitoring and evaluation of the various aspects of a project, service or facility to maximize the probability that standards of quality are being attained by the production process. Quality Assurance cannot absolutely guarantee the production of quality products. The quality department of AFBL is assigned to maintain consistently uniform quality of the material in process and various stages of its manufacturing. Quality assurance department checks the quality of a sample through different test and takes proper action to ensure desired quality. The processes that ensure product quality meets the requirement of customers.

- Base on processes
- Builds quality in

5.2.5 Test perform in QA Department

- Tensile strength
- Tearing strength
- Expiry date
- Flavor taste
- Bursting
- Bottle length & Width

5.2.6 Differences between quality assurance (QA) and quality control (QC)

QA:

- Focus on process.
- Achieved by improving production processes.
- Targeted at the whole business.
- Emphasizes the customer.

QC:

- Focus on outputs.
- Achieved by sampling and checking (inspection).
- Targeted at production activities.
- Emphasizes required standards

5.2.7 Quality standard (ISO)

The International Organization for Standardization (ISO) is an international standard-setting body composed of representatives from various national standards organizations. It is an independent, non-governmental organization, the members of which are the standards organizations of the 164 member countries. It is the world's largest developer of voluntary international standards and facilitates world trade by providing common standards between nations. Over twenty thousand standards cover everything from manufactured products and technology to food safety, agriculture, and healthcare.

The main quality management system of **AFBL** group is:

ISO-9001:

ISO 9001 is the family of standards for quality management. It is maintained by the international organizational for standards for standardization and is administered and is administered by the accreditation and certification bodies.

ISO 9001:2015 specifies requirements for a quality management system when an organization:

- Needs to demonstrate its ability to consistently provide products and services that meet customer and applicable statutory and regulatory requirements.
- Aims to enhance customer satisfaction through the effective application of the system, including processes for improvement of the system and the assurance of conformity. All the requirements of ISO 9001:2015 are generic and are intended to be applicable to any organization, regardless of its type or size, or the products and services it provides.

Halal Food:

Halal certification applies to the food, cosmetic and pharmaceutical sectors and attests that a product is manufactured in full compliance with the precepts of Islamic Law, that it does not include any “forbidden” components, and has in no way been in contact with any substances or objects considered “impure”. An internal company Halal QA committee should exist to regularly review and audit their record keeping for the ordering, receipt and identification of Halal ingredients, production equipment sanitization steps, product packaging, labeling and segregated storage areas. In the Quran, the word halal is contrasted with haram (forbidden). In Islamic jurisprudence this binary opposition was elaborated into a more complex classification known as the five decisions are mandatory, recommended, neutral, reprehensible, and forbidden.



HACCP WCS:

HACCP is a management system in which food safety is addressed through the analysis and control of biological, chemical, and physical hazards from raw material production, procurement and handling, to manufacturing, distribution and consumption of the finished product. Dairy Grade a Voluntary HACCP by using WCS logo. Certification can be beneficial because of increased pressure by customers and consumers, to demonstrate commitment to local authorities and other licensing organizations, to gain independent feedback on the effectiveness of any food safety system, to improve organization’s image.

BSTI:

BSTI formulates and issues Bangladesh Standards (BDS) for products and services inside the country. The products and services which maintain Bangladesh Standards are certified by the institution, and a Certification Mark (CM) license is issued for a validity of three years after factory inspection and standardized testing of the products in its laboratories. Only the licensed manufacturers can use the BSTI certification mark in their product package. The institution has formulated over 4000 standards since its inception. 239 products require mandatory CM license to be produced, distributed and sold inside Bangladesh. According to Bangladesh's import policy, 79 products require mandatory clearance to be imported into the country. The primary activities of BSTI Registration the Bangladesh Standards and Testing Institution (BSTI) are standard services and products.



Introduction of the international unit system of weights and measures and promotion of metrology services. Promotion of quality assurance activities; rendering testing facilities for services and products; preparation, promotion and adoption of national standards. The Institution is also empowered with some regulatory measures in these fields. The Institutional budget is fully supported.

5.3 Time Management in AFBL

Time Management is the process of planning and exercising of conscious control of time spent on specific activities, especially to increase effectiveness, efficiency and productivity. It involves the balancing of various demands upon a person relating to work, social life and family. It also added by a range of skills, tools and techniques used to manage time when accomplishing specific tasks, projects and goals complying with a due date. Initially time management referred to just business or work activities but eventually, the term broadened to include personal activities as well. Mainly AFBL follows some tips as time management:

- Stop multitasking
- Set deadlines
- Prioritize tasks
- Remove distractions
- Keep mind fresh
- Work when most productive
- Set reminders
- Batch your tasks
- Decline additional commitments
- Get more organized
- Get in a routine
- Manage stress
- Delegate tasks
- Break big projects into steps
- Only take on what you can finish
- Get inspired.
- Only focus on what matters.

5.4 Improvement Management System in AFBL

In AFBL they follow these as improvement method:

5S:

It is a workplace org. method that uses a list of five Japanese words. Steps of 5s: Sort, Set in order, shine, standardize & sustain.

5W1H

It is the basic questions whose answers are considered basic in information gathering or problem solving. They are often mentioned in some other workplaces. The method is:

1. Describe the initial situation
2. Determine the key factors
3. Priorities them
4. Propose fitting with importantly and effectively

5.5 Supply Chain Management

Supply chain management (SCM) is the broad range of activities required to plan, control and execute a product's flow, from acquiring raw materials and production through distribution to the final customer, in the most streamlined and cost-effective way possible. In AFBL, this process starts with the sales who done a market survey; after that the marketing team has done the forecasting for an extended period which is revised from time to time. Then the forecasting came to the supply head, who contacted the logistics and the procurement team. The procurement team purchases the products; the raw materials are often globally bought and delivered to the specific country as per requirements. Then the products come to the storage who received it but not being an asset to the Company they are QC passed. After QC passing the raw materials come to the storage and then send to the different production floor as per requisition. After production, the finished goods are stored in the production premises and then sent to the depot after being passed by QC for distribution to the distributor, and then they come to the customer end.

5.6 Supply Chain System in AFBL

A master production schedule (MPS) is a plan for each time period's various commodities to be produced, such as production, personnel, inventories, and so on. Employees may be processing orders from projections or needs from client orders, resulting in separate demand with production schedules. DRP stands for Distribution Requirements Planning. To satisfy demand, DRP focuses on moving goods through the supply chain.

MRP (Material Requirements Planning) is a common supply planning technique that assists organizations, mainly product-based manufacturers, in understanding inventory requirements while balancing supply and demand. It's a computer-based inventory management system that aims to boost corporate productivity. Material needs planning systems are used by businesses to predict raw material amounts and schedule delivery

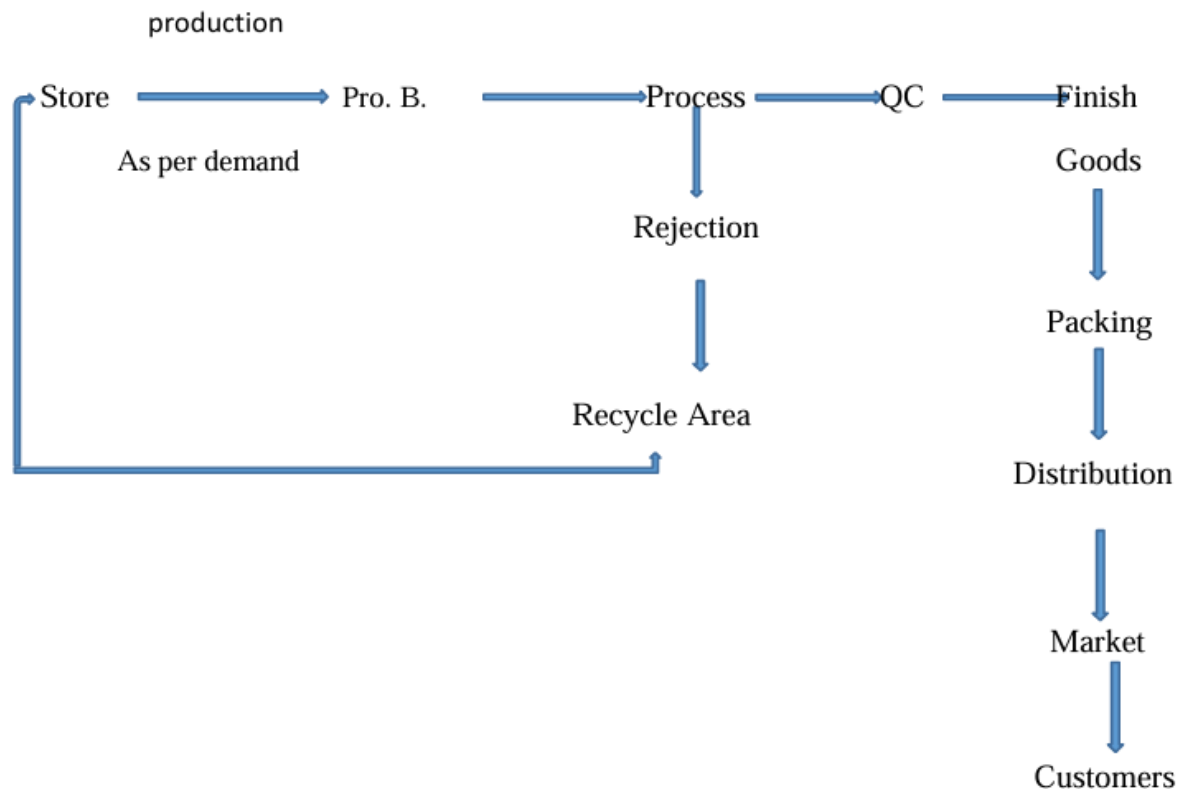


Figure 5.1: AFBL Supply Chain System

CHAPTER SIX

MAINTENANCE

6.1 Maintenance of Machinery

Machine plays very important role in every industry. AFBL also is not separate from this. Since last 15 years there is big boom in industry for automation and ultra-modern, self -sufficient factories. Maintenance of machinery is a procedure by means of which we can maintain active functioning in operation according to the behavior and utility of particular element. Maintenance of machinery in AFBL mainly works one time in every one year. Again it is the process of systematic activities done for keeping the machine or Equipment at the heat level for well run or its proper services.

6.1.1 Maintenance Department

The maintenance Department mainly deal with corrective and preventive action, quality inspection for machines installation, Process related quality of operation qualification, Performance testing. They do check and making SOP!!! The maintenance department works from the factory gate, canteen, the window of the building, road all over the stuff maintenance. They also collect the time gap or how much time the production was off and take action.

6.1.2 Types of maintenance

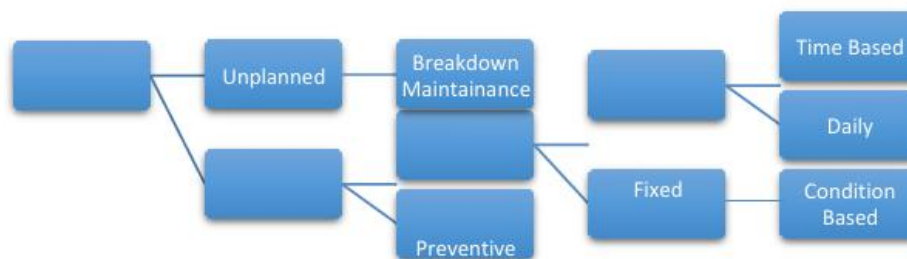


Figure 6.1: Maintenance Types

6.2 Routine for Maintenance

The maintenance department has a specific routine for all of the machines checking and inspection. They maintained a board where they constantly updated the dates of maintenance and the future date of care. Some of the devices need to be held weekly, and some machines maintenance is done with the turnover or changeover. In general, they do weekly, monthly and annual maintenance and make a report.

6.3 Maintenance Procedure

Four types of utility they maintained under maintenance

- Water
- Air
- Power/Electricity
- Gas

6.3.1 Water Purification System

At first, they suck the groundwater through the pump; then the water goes to multigrid sand filter through feed pump where the PH is controlled, then the water goes to activated carbon where the TDS is contained; they try to keep the TDS below 5. After that, if the TDS is still not maintained, it is filtered through the anion and cation tank and reserved in the DM water storage. Then the water is delivered to the production system as per requirement through a transfer pump. And waste water goes to drainage.

6.3.2 Power Maintenance System

The generator cannot run together as the phase sequence is different, so if they run together, an explosion may occur. So they have a unique key system, which allows opening one of the sources at one time; that reduces the chance of happening this severe hazard altogether. They also maintain a 5S system. The total power supply per hour is 540 KVA (**approx.**). They supplied the power to produce as per priority and requirement when a power cut occurs. This is a GDP (Good Documentation Practice) area.

6.3.3 Air Maintenance Procedure

In Company, they need to maintain high pressure and somewhere low. They have an air pressure meter in each production room and necessary places. Such as in DLS, their pressure acceptance limit is 10 to 30 Pa. So they need to compress the environment air and deliver it to the production units. This is done by a vast air compressor that runs 24*7 and has a capacity of 1700 Psi. This air compressor needs to be inspected after each 2200 hour.

6.4 Maintenance of Machinery

Maintenance of Machinery of each and every production unit, office and factory done by the maintenance department. The maintenance department does regular maintenance to machines, check the processes of operations, Quality inspection of installation system. They have electrical, mechanical and plumber team who does the technical operations.

6.5 Manpower set up for maintenance

Mainly in electrical & mechanical section there is a strong manpower arrangement for carrying the production smoothly. The manpower is arranged by following sequence-

Electrical Section & Mechanical Section:

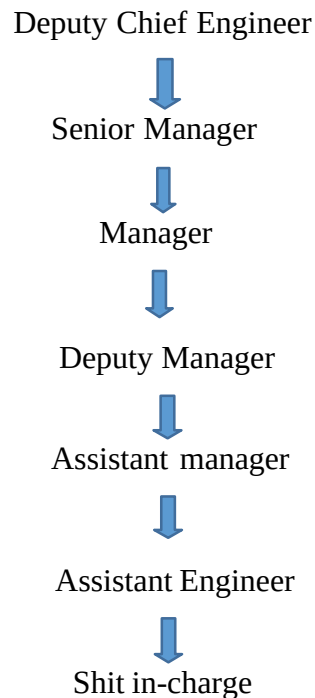


Figure 6.2: Manpower Set up

6.6 Maintenance Procedure

The following aspects should be considered with respect to Machine Maintenance Procedures:

- **Clean and inspect:** Eliminate all dirt and grime on the machine, tighten bolts, lubricate, and find the correct problems.
- **Eliminate problem sources and inaccessible areas:** Correct sources of dirt and grime; check to spatter and develop accessibility for cleaning and lubrication.
- **Draw up cleaning and lubricating standards:** Write standards that will ensure that cleaning, lubrication, and tightening can be done efficiently
- **Conduct The general inspections:** Conduct skills training with inspection manuals and use general inspections to find and right slight abnormalities in the equipment.
- **Conduct Autonomous inspections:** Prepare regular check sheets for autonomous inspections.
- **Carryout the inspections:** Standardize through visual workplace management.

6.7 Maintenance tools/Equipment & their function

All tools, equipment, and vehicles must be properly maintained so that workers are not endangered. Construction regulations require inspections of vehicles, tools, machines, and equipment before use.

Preventive maintenance is the systematic care and protection of tools, equipment, machines, and vehicles in order to keep them in a safe, usable condition, that limits downtime and extends productivity. AFBL is always aware that maintenance tasks themselves are potentially hazardous and can result in injury. The successful maintenance program is:

1. Well organized and scheduled
2. Controls hazards
3. Defines operational procedures
4. Trains key personnel.

The degree of detail to include in AFBL equipment maintenance will depend on the kinds of tools/equipment used. Some construction equipment has very specific inspection and maintenance requirements. Mobile heavy equipment (dozers, loaders, scrapers, etc.) may have different maintenance requirements. Passenger vehicles (company trucks, cars, and vans) may require only basic maintenance. Power tools should be maintained in good working order. This may be limited to ensuring that blades/bits are replaced when needed and that guards or other safety devices are operable and any damaged electrical cords/plugs are repaired or replaced. Damaged or defective equipment/tools should be tagged and removed from service.

AFBL provide maintenance schedules for their equipment. Large companies with a fleet of vehicles/equipment typically have a comprehensive maintenance program due to the capital investment and/or leasing agreements. Smaller companies may lease equipment and maintenance services may be included in the leasing agreement.

Table 6.1: Maintenance Equipments

Tool name	Function
Combination tools (Spanner)	Tightening & Loosening of Nuts & bolts
Socket Ratchet set	Tightening of Nuts & bolts
Hammer	Forging hot metal, riveting, bending, Straightening, peening, stretching and swaging.
Screw driver Made of steel.	Is a hand tool, designed to turn or release screws or bolts.
Steel Tape	To measure the length.
Pliers	To grip anything & cut metal wires.
Hacksaw blade	To cut any metal thing.
Punch	Used to fit any worm out shaft.
Oil Can & Grease Gun	For Oiling & greasing of moving Parts of M/C.
Drill M/C and Drill bit.	For Drilling.
Toolbox	Generally, all necessary maintenance tools are kept in a box it is called tool box. Generally, it is made of steel.

CHAPTER 7

COST ANALYSIS

Costing of product is very important for running of business. **AFBL** puts much importance on costing. Costing of products is made basis on the Raw Material Expenditure, Salary & Wages of Officials & Workers, Distribution & Advertisement Expenses etc. It is determined by accountants with advice & consultancy of executive director.

7.1 Cost Analysis

COVID-19 has made a tremendous impact on human life. Country wise lockdown and maintaining social distancing has disrupted the daily life. As a result, business faced a great challenge to survive for these years. AFBL made a gross sale of (approximately) Tk. 564.1 cr. in 2022 (Can-1) production line. Costs of other raw materials for packaging like Resin, Tetra Packs Foil has increased substantially. The Management reviewed the wages of the workers' according to the existing labor law and increased over the previous due to inflation in food items. This has an adverse impact on the cost of production. AFBL wanted to cover up the costs by increasing sales, but tried to avoid hiking of price of consumer goods. Most of their demandable product price is static in the last fiscal year. In the face of the adverse situations your Company earned Tk.6.81 Cr. operating profit and Tk. 4.55 Cr. net profit, after deducting Workers Profit Participation Fund (WPPF) and Corporate Tax including deferred tax.

The comparatives on the financial performance of the Company for twelve-month period of 2022 are presented below:

Table 7.1: All values are approximate value

Particulars	(Taka)
Revenue	2,641,267,631
Cost of Goods Sold	2,147,164,973
Gross Profit	494,102,658
Operating Expense	296,148,798
Net Profit	68,828,522
Provision Tax	23,249,636
Net Profit After Tax	45,578,886

Statement of Profit or Loss and Other Comprehensive Income

Particulars	(Taka)
Revenue	2,641,267,631
Less: Cost of goods sold	2,147,164,973
Gross Profit	494,102,65
Less: Operating Expense	296,148,798
Administrative expenses	65,769,238
Marketing expenses	93,232,346
Selling Expenses	44,782,891
Distribution expenses	92,364,323
Gross Operating Profit for the Year	197,953,860
Less: Finance Expense	129,794,318
Net Operating Profit for the Year	68,159,542
Other income	4,110,406
Profit before contribution to worker's profit participation fund	72,269,948
Less: Provision for contribution to WPPF & WF	3,441,426
Profit Before income tax	68,828,522
Less: income tax expense	23,249,636
Current Tax	25,518,394
Deferred Tax	2,268,758
Net Profit after income Tax	45,578,657
Total comprehensive income	45,578,886
Earnings Per Share (EPS)	5.70

CHAPTER 8

OTHERS

8.1 Shipment and Delivery System

Shipment system is very important for **AFBL** as they import a lot of raw materials from outside of the country. This is done mainly by the Chittagong Sea port and air cargo. Any of the raw material which is imported from outside has come to the factory gate and receive by storage with invoice. Then they sent to the quality lab for further testing. As long as they get the QC passed, they are considered as a company asset. Finished Goods they shipment to the distributor.

8.2 Industrial Compliance and Safety

Safety culture is the most amazing stuff in **AFBL**. Everywhere every time one is reminded by their safety, PPE, monthly golden rule of ergonomics, safety motivational quotes, poster, mirror etc. Their one of the mottoes is "Safety first, Safety Last, Safety Always". The first thing we need to maintain after coming is maintaining safety shoes. As they ensure the safety of employee and employer. In the factory premises there is a standby ambulance and a doctor. They had fire hydrant system in their organization. As we visited in different production floor employees are maintaining their appropriate safety precautions, PPE. MSDS is presented in every unit and they review it periodically, in general annually. Manual material handling procedures from ergonomics or human factor is followed strictly. In production department these ergonomically designed gestures and postures are presented everywhere. They have walking bar rails in walking ways and yellow marking on the roads what symbolizes the road crossing and walking ways. In storage the materials keep 10 inch aside of the wall as this is necessary for safety issue.

8.3 Treatment Plant

AFBL prioritizes environmental sustainability where we think green and work green. Our motto is to achieve a Sustainable Green Planet someday. To protect and conserve this beautiful mother earth we are taking few green initiatives into account.

- Recycle
- ETP
- WTP

Furthermore, we have also generated 2 ETP & 1 WTP to ensure safe factory.

8.3.1 The Components of Recycling Process in AFBL

Wastage disposal, community forestation program to balance out carbon dioxide amount in the air, heat recovery boiler for efficient reuse of the heat from production and we've also switched to fossil fuel and powered trucks to reduce air pollution. Additionally, we utilize daylight with skylight roof and CFL Bulbs at all our offices and factories to reduce power consumption.

8.3.2 Machine Used in Recycling Process

- Requester machine
- Crusher machine
- Washing machine

8.3.3 Recycle process in AFBL

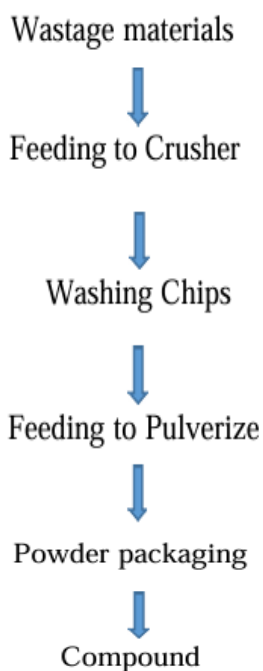


Figure 8.1: AFBL Recycle Process

8.3.4 ETP

Effluent treatment plant, also known as ETP is a waste water treatment process (WTP) that is used to treat waste water. AFBL use this ETP, where waste is treated by some process. They have 2 ETP plant for treating their plant waste water. Industrial effluents contain various materials, depending on the industry. Some effluents contain oils and grease, and some contain toxic materials (e.g., cyanide). Effluents from food and beverage factories contain degradable organic pollutants. Since industrial waste water contains a diversity of impurities and therefore specific treatment technology called ETP is required. The ETP Plant works at various levels and involves

various physical, chemical, biological and membrane processes to treat waste water from different industrial sectors like chemicals, drugs, pharmaceutical, refineries, dairy, ready mix plants & textile etc.

8.3.5 Process flow chart of ETP

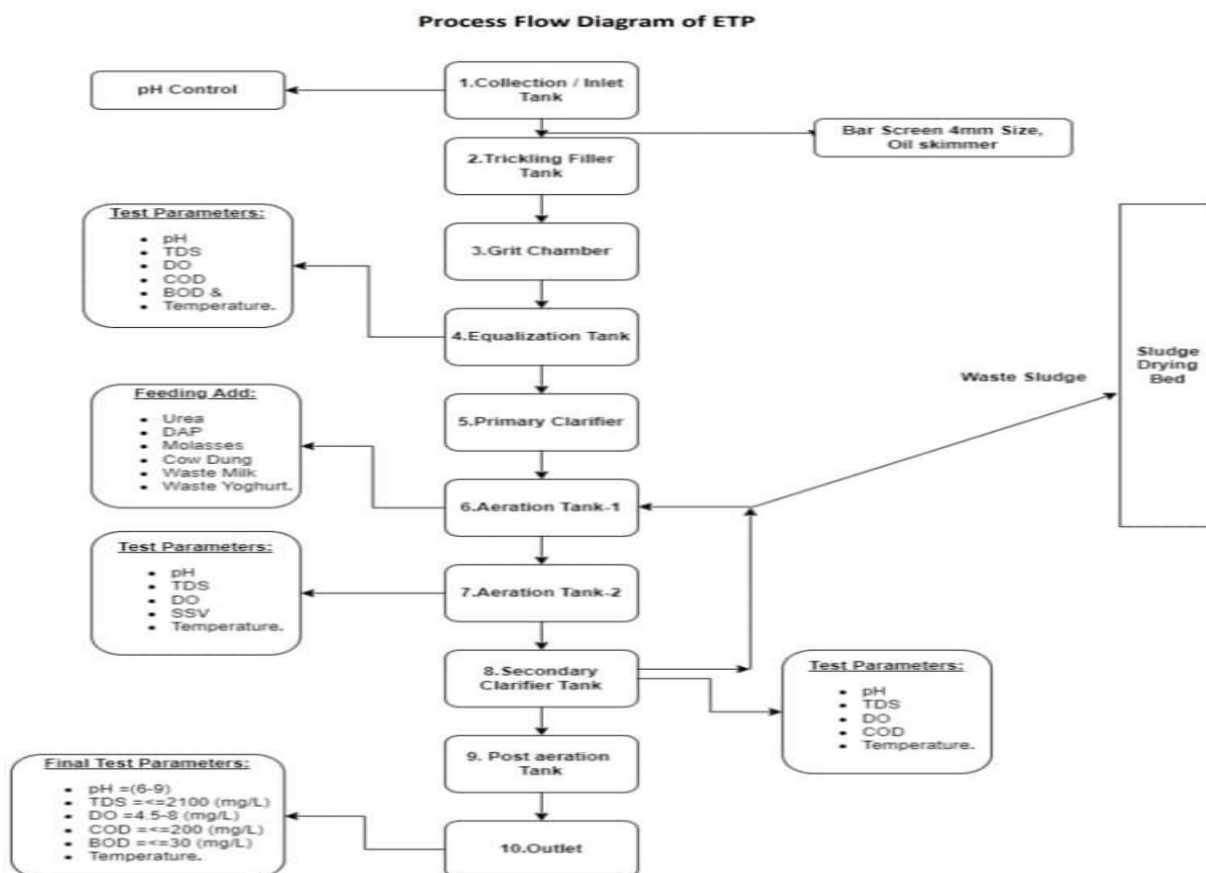


Figure 8.2: ETP Process

WTP

Water is one of the most important utilities in **AFBL** in every sector. Without water this industry cannot imagine a single production. But for various production these water sources cannot be used directly in boilers as such. The objective of water treatment plant is to produce the boiler feed water so that there shall be:

- No Scale Formation
- No Corrosion
- No Fouling

The treatment water is called “**De-Mineralized**” water & the plant where it is treated is called “**Water Treatment Plant**”.

8.3.6 Machine in WTP

- Lamicorn Filter
- Sediment Filter

8.3.7 Process Flow chart of WTP

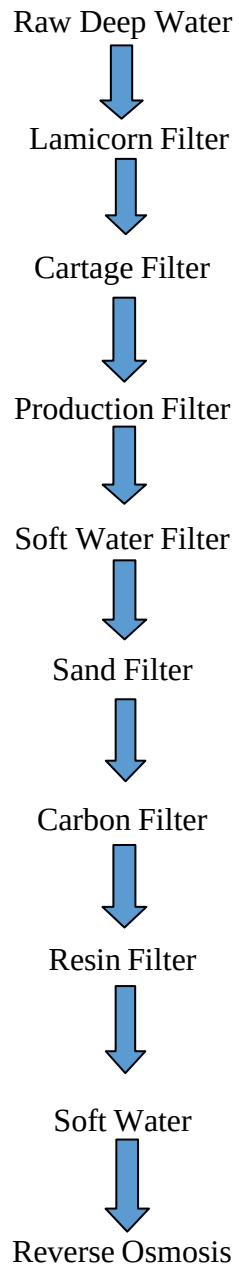


Figure 8.3: WTP Process

CHAPTER 9

IMPACT OF THE INTERNSHIP

Internship at **AFBL** helps us to know about the process performed in an industry. It makes us strong enough to perform our duties in job properly. It also helps us to develop our skill through different activities. In this training at AFBL we have gained many things.

We are fortunate enough, because we have merged our theoretical knowledge with the professional experience in this 1-month internship in such a well reputed leading company in Bangladesh. This training gave us an actual picture about man, machine, material, methods and market of (IPE). We have earned the direct knowledge about the raw materials, actual running condition of the machines, works technologist and administration.

CHAPTER 10

CONCLUSION

This internship has been a very useful experience for us. We can safely say that our understanding of the job environment has increased greatly. We realized that we could have completed the work earlier than we did. Also, the technical parts of the job were a bit flawed and we were asked multiple times to correct it. The two main things that we learned after our experience in this time frame is importance of Time management and being Self- Motivated.

AFBL is a well-maintained Industry with modern machineries, technologies, good management with 100% compliance, whose main driven force is its human spirit which has marched towards its glory of success. Use of modern equipment's in manufacturing, looking after the welfare of the workers, good management are the key to success of a company which are well maintained.

In Industrial & Production Engineering (IPE) Education practical knowledge is often more important than theoretical knowledge. Theoretical background is not sufficient so, industrial training is an essential part of study to make a technologist technically sound in this field. Industrial training provides us that opportunity to gather practical knowledge. This Industrial training increases our thought a Total Industrial Technology.

During this internship period, we got all kinds of support & co-operation from the Authority to gather practical knowledge on the Industrial & Production Engineering (IPE). All stuffs & officers were very sincere & devoted to their duties to achieve their goal. And, we're Thankful to the HR Department of AFBL for arranging our internship program and providing facilities and atmosphere to learn. With their guidance we passed a memorable Time with AFBL during our learning time.

Finally, we would like to express our gratitude from the core of our heart towards all the members of AFBL and our respected teachers to guide us in every feasible way. We tried our best to gather all necessary Information, but within this short period it is quite impossible to achieve 100% success.

Surely This Professional Experience will help us set the first brick to build our Career in the desired sector. We believe with all these, the Experience of the Industrial Attachment would help us in our professional life as an Industrial & Production Engineer.

Appendix

Certificates


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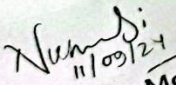
Certificate for Industrial Attachment

This is to certify that Jahidul Islam, S/O- Aleya Begum, bearing Student ID: JP-2004015, session 2019-2020, pursuing his Bsc In Industrial & Production Engineering at National Institute of Textile Engineering And Research has successfully completed 01 (One) month (From 15th May 2024 to 12th Jun 2024) Internship program at Akij Food & Beverage Ltd.

Jahidul Islam displayed professional traits during his internship period and managed to complete all assigned tasks as requested. He was hardworking, dedicated, and committed. It was a pleasure having him with us in this short period.

We wish his every success in life.

For Akij Food & Beverage Ltd.,


Md. Nurun Nabi 11/09/24
Asst. Sr. Manager-ER-HR
Assistant Senior Manager
IIR & Admin

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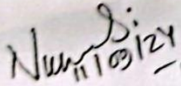
Certificate for Industrial Attachment

This is to certify that Sadia Islam Hridita, D/O- Md. Saiful Islam, bearing Student ID: JP-2004070, session 2019-2020, pursuing her Bsc In Industrial & Production Engineering at National Institute of Textile Engineering And Research has successfully completed 01 (One) month (From 15th May 2024 to 12th Jun 2024) Internship program at Akij Food & Beverage Ltd.

Sadia Islam Hridita displayed professional traits during her internship period and managed to complete all assigned tasks as requested. She was hardworking, dedicated, and committed. It was a pleasure having her with us in this short period.

We wish her every success in life.

For Akij Food & Beverage Ltd.,


Md. Nurun Nabi
Assistant Senior Manager
HR & Admin
Asst. Sr. Manager-ER-HR
Akij Food & Beverage Ltd.

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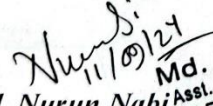
Certificate for Industrial Attachment

This is to certify that Ahwas Al Hossain, S/O- Musfiqua Binte Sayeed, bearing Student ID: JP-2004075, session 2019-2020, pursuing his Bsc In Industrial & Production Engineering at National Institute of Textile Engineering And Research has successfully completed 01 (One) month (From 15th May 2024 to 12th Jun 2024) Internship program at Akij Food & Beverage Ltd.

Ahwas Al Hossain displayed professional traits during his internship period and managed to complete all assigned tasks as requested. He was hardworking, dedicated, and committed. It was a pleasure having him with us in this short period.

We wish his every success in life.

For Akij Food & Beverage Ltd.,


Md. Nurun Nabi
Asst. Sr. Manager-ER-HR
Akij Food & Beverage Ltd.
Assistant Senior Manager
IIR & Admin

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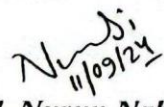
Certificate for Industrial Attachment

This is to certify that Adnan Khan, S/O- Shajhan Khan, bearing Student ID: JP-2004090, session 2019-2020, pursuing his Bsc In Industrial & Production Engineering at National Institute of Textile Engineering And Research has successfully completed 01 (One) month (From 15th May 2024 to 12th Jun 2024) Internship program at Akij Food & Beverage Ltd.

Adnan Khan displayed professional traits during his internship period and managed to complete all assigned tasks as requested. He was hardworking, dedicated, and committed. It was a pleasure having him with us in this short period.

We wish his every success in life.

For Akij Food & Beverage Ltd.,


Md. Nurun Nabi
Aast. Sr. Manager-ER-HR
Assistant Senior Manager
IIR & Admin
Akij Food & Beverage Ltd.

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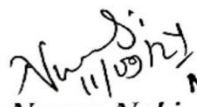
Certificate for Industrial Attachment

This is to certify that Aditya Roy Chy, S/O- Jnanatosh Chy, bearing Student ID: JP-2004108, session 2019-2020, pursuing his Bsc In Industrial & Production Engineering at National Institute of Textile Engineering And Research has successfully completed 01 (One) month (From 15th May 2024 to 12th Jun 2024) Internship program at Akij Food & Beverage Ltd.

Aditya Roy Chy displayed professional traits during his internship period and managed to complete all assigned tasks as requested. He was hardworking, dedicated, and committed. It was a pleasure having him with us in this short period.

We wish him every success in life.

For Akij Food & Beverage Ltd.,


Md. Nurun Nabi
Assistant Senior Manager
HR & Admin

Md. Nurun Nabi
Asst. Sr. Manager-ER-HR
Akij Food & Beverage Ltd.

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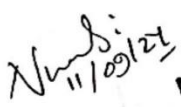
Certificate for Industrial Attachment

This is to certify that Shuvro Biswas, S/O- Mita Biswas, bearing Student ID: JP-2004413, session 2019-2020, pursuing his Bsc In Industrial & Production Engineering at National Institute of Textile Engineering And Research has successfully completed 01 (One) month (From 15th May 2024 to 12th Jun 2024) Internship program at Akij Food & Beverage Ltd.

Shuvro Biswas displayed professional traits during his internship period and managed to complete all assigned tasks as requested. He was hardworking, dedicated, and committed. It was a pleasure having him with us in this short period.

We wish his every success in life.

For Akij Food & Beverage Ltd.,


Md. Nurun Nabi
Asst. Sr. Manager-ER-HR
Akij Food & Beverage Ltd.
Assistant Senior Manager
HR & Admin

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